Chemical Engineering Journal 525 (2025) 170195

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

Review

Surface morphology engineering of triboelectric nanogenerators for performance enhancement

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ARTICLE INFO

Keywords:
Triboelectric nanogenerator
Surface morphology engineering
Performance enhancement
Interfacial engineering
Wearable electronics

ABSTRACT

Triboelectric nanogenerators (TENGs) have emerged as a transformative technology for harvesting ambient mechanical energy. However, their practical application is often constrained by their electrical output performance. Surface morphology engineering has been identified as a critical strategy to overcome this limitation by maximizing charge generation and transfer at the triboelectric interface. This review provides a comprehensive and systematic overview of recent advances in enhancing TENG performance through the deliberate control of surface morphology, highlighting the central role of chemical engineering principles. We critically examine the roles of various materials including metals, ceramics, and polymers, and elucidate how their surface properties influence device output. Furthermore, we categorize and analyze key fabrication processes (e.g., chemical, physical, and hybrid) and structural designs (e.g., micro, nano, and hierarchical) that enable precise morphological control. Finally, we discuss emerging research trends, identify persistent challenges, and offer future prospects for the development of high-performance, mechanically robust TENGs for next-generation applications in wearable electronics, self-powered sensors, and the Internet of Things (IoT). This review aims to provide a strategic guide, from a chemical engineering perspective, for the rational design of advanced TENGs based on a fundamental understanding of molecular-level interfacial engineering.

1. Introduction

In recent years, mounting concerns regarding global energy depletion, carbon emissions, and environmental degradation have intensified both academic and industrial interest in sustainable and environmentally responsible energy technologies [1–5]. Conventional battery-based power systems exhibit inherent limitations, including restricted operational lifetimes, recurring recharging requirements, and adverse environmental impacts, which compromise their suitability for next-generation electronic devices [6,7]. These limitations are particularly critical in wearable electronics and distributed IoT sensors, which demand enhanced flexibility, durability, and autonomous operation. In such applications, batteries or wired power sources not only impose physical constraints but also experience performance degradation under diverse environmental conditions, ultimately reducing overall system efficiency and practical viability.

Consequently, interest has intensified in novel energy sources capable of continuous and autonomous power generation. Among these technologies, energy harvesting systems, which directly convert ambient energy into electrical power, have attracted considerable attention. Principal mechanisms encompass photovoltaic, thermoelectric, piezoelectric, and electrostatic approaches, each offering distinct advantages and limitations depending upon specific applications [8–11]. Among these technologies, the TENG, a relatively recent innovation, has emerged as a promising approach for converting diverse forms of environmental mechanical energy into electricity. Since its initial demonstration in 2012, TENG technology has advanced rapidly due to its simple architecture, lightweight characteristics, material versatility, and cost-effective fabrication methodologies. These devices demonstrate considerable potential for applications in wearable sensors,

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https://doi.org/10.1016/j.cej.2025.170195
Received 4 September 2025; Received in revised form 2 October 2025; Accepted 26 October 2025
Available online 2 November 2025
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self-powered electronics, IoT systems, and integration with compact energy storage devices [12–18].

TENG operation relies upon two fundamental physical principles: triboelectrification and electrostatic induction. The properties of contact surfaces and their structural design critically influence charge generation and energy conversion efficiency. Engineering the surface morphology of triboelectric interfaces can enhance charge density and electrostatic capacity by enlarging effective contact areas, improving output performance. Surface morphology can be modulated through various approaches ranging from micro- and nanoscale patterning and structural design to chemical functionalization and hybrid processing techniques [19–22]. These strategies not only enhance electrical performance but also improve mechanical durability and operational reliability, enabling broader TENG applications in integrated design frameworks.

In addition to these material and structural innovations, recent surface engineering advances have notably increased the output per unit area (cm^2) of TENG devices. This improvement in areal output not only enhances overall device power density but also offers practical advantages, including scalability, reduced fabrication time and material consumption, and expanded applicability for flexible or large-area devices [23]. These aspects further underline the technological significance and



Fig. 1. Overview of surface morphology-engineered TENG.
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practical benefits of precise surface morphology control in TENG design.

Materials employed in TENGs are typically classified into three primary categories: metals, polymers, and ceramics. Each category is selected according to its function as electrodes, triboelectric interfaces, or functional components. Polymers are widely utilized as triboelectric layer due to their inherent dielectric and mechanical properties such as high dielectric constant, charge tendency modification, flexibility, and excellent processability. Moreover, polymers with the low dielectric constant are used as encapsulation to protect charge leakage [24]. Ceramic materials provide high mechanical strength and thermal stability, with some exhibiting intrinsic dielectric or piezoelectric properties that render them advantageous for developing multifunctional energy harvesting devices [25].

Surface treatment and structural design strategies for TENGs can be systematically categorized into physical, chemical, and hybrid approaches [26–28]. The central paradigm involves maximizing electrostatic coupling at triboelectric interfaces through deliberate micro- and nanoscale patterning and construction of hierarchical surface architectures. Recent studies have demonstrated significant enhancements in output voltage and current density using methodologies including micro-patterned surface fabrication, plasma-assisted modification, chemical doping, and synergistic hybridization of these techniques [29–31]. Accordingly, the engineering of surface morphology for TENG performance optimization has evolved into a highly interdisciplinary field that integrates structural design principles, advanced surface modification techniques, and interfacial electrical regulation, transcending the limitations of simple material selection.

This review provides a comprehensive examination of recent research trends and technological advances aimed at enhancing TENG performance through surface morphology engineering, and also dedicated to future prospects, discussing overlooked challenges that must be addressed to further drive the development and practical applications of high-performance TENGs. We systematically examine strategies involving triboelectric materials, surface structure design, and fabrication processes, with emphasis on the properties of metallic, polymeric, and ceramic materials; the design of micro-, nano-, and hierarchical structures; and the processing methods employed to realize them, including lithography, plasma treatment, and chemical doping (Fig. 1) [32–38]. In particular, this review emphasizes that the core of this progress lies in the sophisticated application of chemical engineering. Molecular-level strategies such as controlled polymer synthesis, targeted interfacial chemical modification, precision doping, and hybrid chemical-physical process design are highlighted as critical enablers of surface morphology control. These approaches, rooted in fundamental chemical engineering disciplines, not only improve triboelectric charge density and interfacial stability but also provide rational chemical pathways that facilitate eco-friendly material design, scalable processing, and enhanced functional integration. This overview establishes a foundation for practical deployment of TENG technologies while highlighting future directions and the potential of TENGs as high-performance self-powered energy sources for diverse industrial applications, including smart sensors, wearable devices, and intelligent healthcare systems.

1.1. Fundamental theory of TENG performance

The performance of a TENG is dominated by the coupling of contact electrification and electrostatic induction. The foundational metric to indicate the potential output of a TENG is the surface charge density (σ), which generates upon two different layer contact. The key performance outputs including the open-circuit voltage (V_{OC}) and the short-circuit transferred charge (Q_{SC}) are directly proportional to this charge density. The relationships are described by the following fundamental equations for a parallel-plate TENG model:

$$V_{OC} = \sigma x/\varepsilon_0 \tag{1}$$

$$Q_{SC} = \sigma \mathrm{A} \tag{2}$$

Here, x is the separation distance between the two triboelectric layers, A is the contact area, and ε_0 is the vacuum permittivity. These equations clearly indicate that maximizing the surface charge density (σ) is the most critical strategy for enhancing TENG performance. While material selection is paramount, engineering the surface morphology at the micro- and nano-scales provides a powerful and universally applicable method to increase σ and overall device efficiency. This is achieved through a synergistic interplay of enhanced contact electrification, improved charge trapping and retention, and modified dielectric polarization. When two different materials come into contact, electrons migrate from one surface to another based on differences in electron affinity and work function, resulting in oppositely charged surfaces. Materials with higher work function indicate more electronegative readily accept electrons from materials with lower work function. However, the subsequent motion of these charges enables useful electrical power generation in a TENG not just charge accumulation. When the charged surfaces are separated, mechanical work is converted into electrical energy via electrostatic induction: the changing geometric configuration induces a voltage and drives current through an external circuit (Fig. 2) [39,40].

1.2. Morphology–performance trade-offs and failure mechanisms in TENGs

Although parameters such as humidity and vibration frequency can influence TENG power output, surface charge density remains the primary figure of merit for evaluating material performance in TENGs. The triboelectric charge density can be enhanced by increasing the contact area. With the limited device size, surface morphological engineering is the initial consideration to increase the effective contact surfaces. Optimal materials selection is straightforward, expanding the effective contact area in a fixed device size is challenging. To overcome this, designers use micro- and nano-scale surface patterns to maximize real contact area.

The enhancement of triboelectric performance with increased surface roughness can be rationalized through fundamental relations [41]. The maximum capacitance of a triboelectric junction can be represented as:

$$C_{max} = \varepsilon_0 \times S\frac{\varepsilon_r}{d} \tag{3}$$

where ε_0 is the vacuum permittivity, ε_r is the relative dielectric constant of the triboelectric layer, S is the effective contact area, and d is the effective separation distance between the layers. With increasing surface roughness, the real contact area S enlarges beyond the apparent geometric area, thereby elevating the capacitance. The open circuit voltage as:

$$V_{OC} \approx Q/C \tag{4}$$

where Q represents the triboelectric charge, it follows that V_{OC} is inversely proportional to capacitance for a given charge density. However, roughened or nanostructured surfaces simultaneously enhance Q by increasing localized contact electrification sites and charge trapping states. Thus, the net effect of increasing surface roughness is a coupled rise in both Q and C, leading to a scaling relation where V_{OC} improves as long as the enhancement in charge density outpaces the increase in capacitance. This interplay provides a general design principle: optimizing surface morphology to maximize effective contact area without excessively increasing capacitance is crucial for achieving higher output voltages in TENGs.

Surface roughening, nanostructuring, and hierarchical design are widely recognized for enhancing the electrical output of TENGs, but they also introduce a complex set of trade-offs in terms of durability,

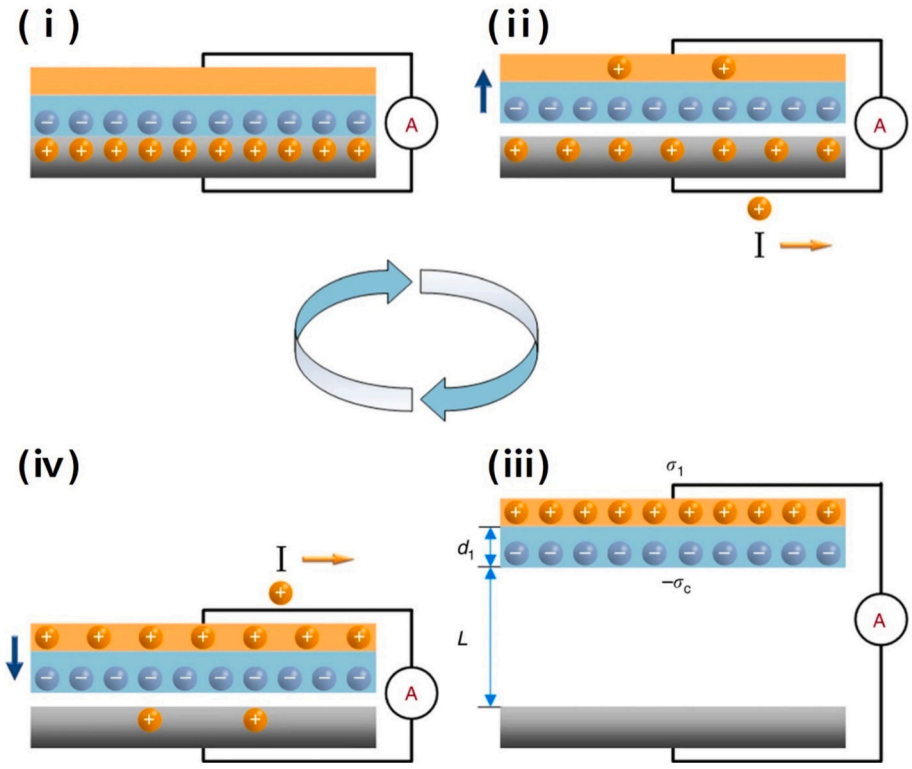


Fig. 2. Schematic illustration of TENG working principle. (i) Full contact of triboelectric layer and substrate plane with pressure, (ii) releasing state with pressure decrease, (iii) released and neutralized substrate with specific distant (*L*), and (iv) approaching state of both layer. Reproduced with permission [40]. Copyright 2020, Springer Nature.

cost, humidity stability, and scalability [42].

Micro-, nano-, and hierarchical morphologies can significantly increase the surface area and effective friction between triboelectric materials, producing higher charge generation and power output. Microstructured surfaces, such as micro-pyramids, generally provide robust mechanical stability and performance enhancement, with relatively better wear resistance under mechanical cycling. Nanostructures such as nanofibers and nanopillars can enhance output by allowing accurate control over local contact electrification. However, their small feature sizes and high surface energy make them susceptible to mechanical wear over time. Hierarchical designs, which combine micro- and nano-scale features, can synergistically improve output, but fabrication complexity and durability concerns escalate, with multi-scale interfaces often at increased risk of delamination or microcrack formation with time [43].

The presence of nanostructured or highly roughened surfaces can make TENGs more susceptible to performance degradation in humid environments. Water adsorption—especially prominent on nano-porous and high-surface-area materials can screen triboelectric charges and form thin conductive films, rapidly reducing output and stability. While hierarchical and nanostructured TENGs may outperform flat designs at moderate humidity, their advantage diminishes or reverses under severe moisture exposure unless special materials or surface treatments are employed [44].

Microstructuring approaches, like mold replication or embossing, are generally more scalable and cost-effective for large-area fabrication. Nanostructuring often requires more sophisticated methods such as electrospinning, nanoimprinting, or complex chemical synthesis, which raise production costs and present challenges for industrial-scale manufacturing. Hierarchical designs demand multi-step fabrication and precise alignment of different layers, further compounding scalability issues and material expense [45].

Wear resistance is generally highest for microstructured TENGs due to their bulkier features and lower susceptibility to mechanical damage. Nanostructures, while excellent for charge generation, suffer rapid decline after repeated mechanical cycling, as the fine features are prone to abrasion and contamination. Hierarchical structures balance output enhancement with intermediate wear resistance, but maintaining structural integrity over thousands of cycles remains a challenge, requiring advanced materials design and robust morphology engineering [46].

In addition to mechanical wear and humidity sensitivity, charge decay remains a critical failure mechanism across morphology designs. Trapped charges at rough or nanostructured interfaces are vulnerable to leakage through surface states, adsorbed water layers, or interfacial defects, resulting in progressive output decline even without visible structural damage. This decay is less pronounced in microstructured morphologies with smoother interfaces, but more severe in nanostructured and porous surfaces where abundant defect sites and high surface energy accelerate charge dissipation. Hierarchical structures exhibit intermediate stability, yet their multiple interfacial boundaries can facilitate charge migration and loss over extended operation [47].

Generally, these comparisons indicate that microstructured morphologies offer the best durability and lowest charge loss, but at the expense of limited enhancement in output. Nanostructured surfaces maximize charge generation and sensitivity but suffer from rapid degradation under cyclic wear, humidity, and charge retention challenges. Hierarchical morphologies achieve superior electrical performance by combining multi-scale effects, but they introduce greater fabrication complexity and long-term risks of delamination or charge decay. Therefore, the choice of morphology is application-dependent: microstructures for robust and scalable harvesters, nanostructures for high-sensitivity but short-lifetime sensors, and hierarchical designs where peak output outweighs fabrication and stability concerns. To provide a clearer comparison of electrical performance and durability across different surface morphology strategies, representative results

from reported studies are summarized in Table 1.

2. Materials

The selection and optimization of materials constitute the fundamental basis for achieving high-performance TENGs through surface morphology engineering. From a chemical engineering perspective, this involves mastering the process-structure-property relationship, where specific synthesis and processing routes are chosen to create desired surface characteristics. While the triboelectric effect depends on the inherent properties of contacting materials, their surface characteristics, engineered through controlled chemical and physical processes, play an equally critical role in determining charge generation efficiency and device stability [52–54]. This section systematically examines how different material categories (e.g., metals, ceramics, and polymers) can be engineered at the surface level to enhance triboelectric performance, with particular emphasis on the relationship between chemically modulated material properties, surface morphology, and electrical output characteristics.

2.1. Metallic materials

Among the various material categories employed in TENG fabrication, metallic materials serve dual functions as both conductive electrodes and active triboelectric surfaces, with their surface morphology playing a decisive role in charge generation and collection efficiency [55,56]. Xia et al. investigated the influence of surface morphology and material composition on triboelectric output performance by testing different metal-based triboelectric layers including Fe, Co, Cr, Ni, and Al foils, comparing them to a FeCoCrNiAl alloy powder layer (Fig. 3a–c) [32]. The planar metal foils demonstrated relatively low performance, primarily attributable to limited effective contact areas. In contrast, the FC-TENG utilizing microspherical FeCoCrNiAl particles exhibited substantially enhanced electrical performance, achieving an open-circuit voltage (V_{OC}) of 1103.02 V, short-circuit current (I_{SC}) of 37.09 μA, and short-circuit charge (Q_{SC}) of 24.15 nC. These outputs exceeded the best-performing pure metals by factors of at least 4.86, 3.57, and 2.76, respectively. Such improvements can be attributed not only to the material's inherent properties as a high-entropy alloy but more significantly to its rough and granular surface structure, a direct result of particle and powder engineering, a key domain of chemical engineering. It increases contact area and facilitates charge trapping in localized regions. Meng et al. conducted systematic investigations of electrical contact properties at metal-polymer junctions and their influence on direct current (DC) output characteristics of TENGs (Fig. 3d–e) [57]. For the Ni-PEDOT-Ni configuration, current-voltage (I-V) curves displayed linear profiles, reflecting conventional ohmic contact behavior. Conversely, the Al-PEDOT-Ni arrangement exhibited distinct rectifying effects, attributed to dynamic Schottky contact formation at the Al-PEDOT interface. This interfacial contact promoted separation of nonequilibrium electron-hole pairs generated during triboelectric processes, assisted by built-in electric fields. A single DC generator based on this triboelectric effect can output a voltage of approximately 0.45 to 0.70 V and a current of approximately 2.5 to 10 μA. The Al-PEDOT-Al design produced very low current densities, emphasizing that both selection and pairing of metals strongly influence charge generation and transport performance. These results highlight that metals in TENGs function beyond mere electrodes, they actively influence energy conversion efficiency through the chemical engineering of interfacial electronic structures and control of electrical barrier formation. Furthermore, tailored surface morphology of PEDOT-coated porous fiber substrates improved adhesion with metal layers, stabilizing Schottky contacts and reducing output degradation under long-term sliding and repeated bending cycles. Khan et al. reported the application of a core chemical engineering process chemical vapor deposition (CVD) synthesis of large-scale three-dimensional Cu_2Te leaf-like structures (Fig. 3f) [58]. The vertically aligned Cu_2Te leaf-like structures considerably increased surface roughness and effective contact areas. When integrated at 4 wt% into polyvinylidene fluoride (PVDF) matrices, the Cu_2Te structures induced β-phase crystallization, improving polarization and charge retention capabilities. The optimized TENG structure exhibited V_{OC} of 170 V and I_{SC} of 32 μA at 40 MΩ load resistance. Zhang et al. reported strategies for improving TENG performance through surface nanostructure formation in metal conductive layers (Fig. 3g–i) [59]. Silver was sputtered onto polypropylene (PP) nanowires (approximately 200 nm) to form nanostructured conductive layers, followed by spin-coating nylon layers to fabricate positively charged triboelectrodes. Scanning electron microscopy analysis revealed that nanowire structures were maintained after Ag layer deposition, with surfaces becoming relatively planar after nylon coating. TENGs based on nanostructured conductive layers exhibited increases in I_{SC} from 26.7 to 37.2 μA and V_{OC} from 348.8 to 414.1 V compared to planar conductive layer-based devices. These improvements result from nanostructures increasing effective surface areas of Ag films, expanding areas available for induced charge generation, generating more induced charges and enhancing output performance. The investigations of metallic materials demonstrate that surface nanostructuring and alloy composition, often achieved through scalable chemical engineering processes like CVD and alloy formulation, significantly outperform planar metallic surfaces, with performance enhancements directly attributable to increased effective contact areas and optimized interfacial electronic properties [60,61].

Table 1
Summary of TENG output characteristics and durability by surface morphology.

Surface morphology engineering	V_{OC} (V)	I_{SC} (μA)	Power density (W/m^2)	Durability (cycles)	Ref
P(VDF-TrFE) quenched film dielectric with charge excitation	246	122	6.67	>324,000	[48]
Multilayer & 3D structuring	145	8.90	1.10	>1200	[49]
Interface chemical modification	398	4.60	0.453	>10,000	[50]
Surface roughness & microhardness	222.8	53	4.256	>26,000	[51]

$$V_{OC} = \sigma x/\varepsilon_0 \quad (1)$$

$$Q_{SC} = \sigma A \quad (2)$$

$$C_{max} = \varepsilon_0 \times S\frac{\varepsilon_r}{d} \quad (3)$$

$$V_{OC} \approx Q/C \quad (4)$$

2.2. Ceramic materials

Chen et al. developed ZnO nanorods (NRs) arrays active layer for improved surface area and charges with Ga doping and achieved improved output performance [62]. Ga doping in ZnO NRs alters their morphology, leading to rougher surfaces, sharper top facets, and a more random alignment of the nanorods (Fig. 4a–b). The Ga-doped ZnO NRs arrays were grown by hydrothermal method with the ZnO seed and gallium nitrate hydrate. This process enables the tunability of charge transfer ability without hybridization with the addition of other materials. Ga doping changes the free electron concentrations up to 1.80×10^{18} cm^{-3} enhanced from 4.24×10^{16} of un-doped ZnO rods. Moreover, output voltage significantly increases from ~2 V to ~27 V by 13 times, whereas the output current density exhibits more dramatic 90-fold enhancement from ~1.32×10^{-9} to ~1.19×10^{-7} A/cm^2. Correspondingly, the tribo-charge density rises from ~1.25×10^{-10} to ~1.91

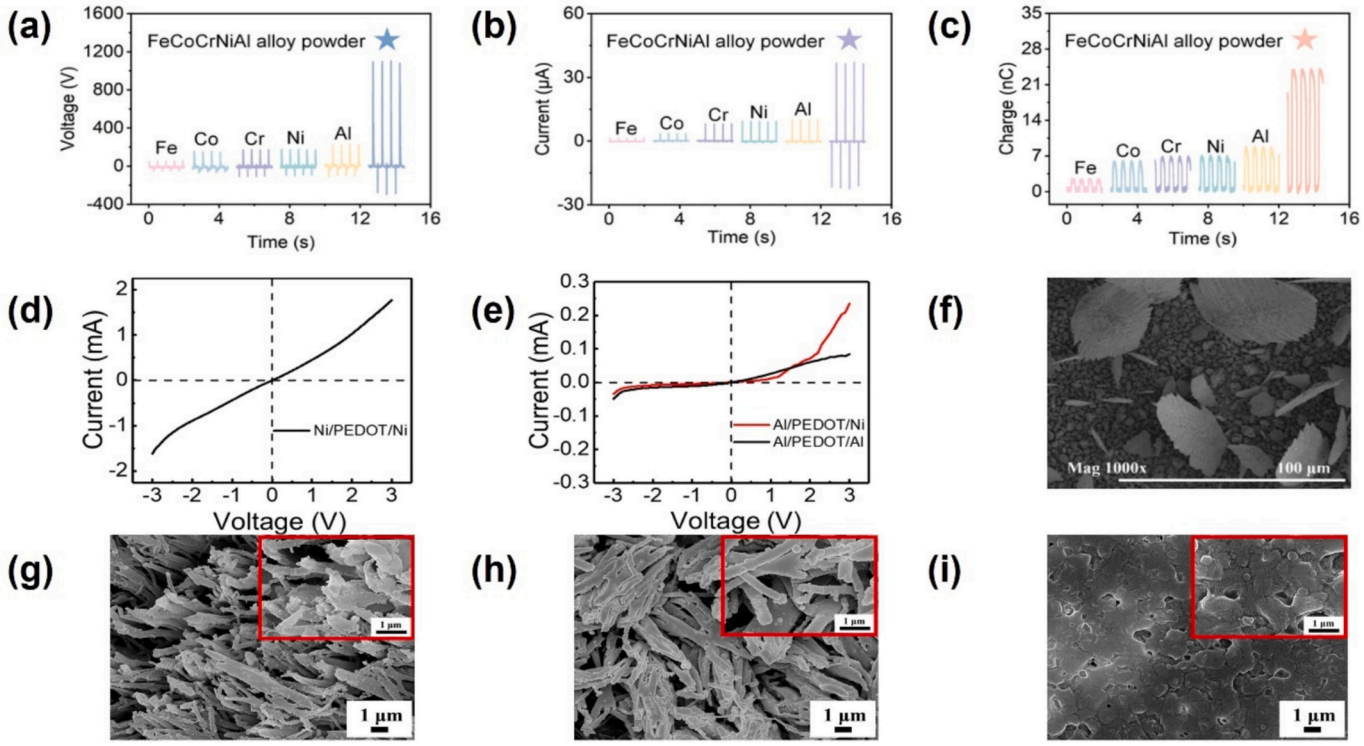


Fig. 3. Electrical properties of metal-modulated TENG; (a) V_{oc}, (b) I_{sc}, and (c) Q_{sc}. Reproduced with permission [32]. Copyright 2025, John Wiley and Sons. I-V curve of (d) Ni/PEDOT/Ni-structured and (e) Al/PEDOT/metal-structured TENG under a 5 N pressure. Reproduced with permission [57]. Copyright 2021, American Chemical Society. (f) EDS analysis of Cu_2Te leaf structure for TENG electrodes. Reproduced with permission [58]. Copyright 2025, Elsevier B.V. SEM images of PP nanowires (g) as-deposited, (h) after silver sputtering, and (i) after nylon spin-coating, which is utilized to TENG active layer.. Reproduced with permission [59]. Copyright 2021, American Chemical Society.

$\times 10^{-9}$ Q/cm^2 by 15 times (Fig. 4c).

However, the pure ceramic-based TENG has confront with the lack of mechanical properties under repeated mechanical stress including low flexibility, stretchability, and elasticity. These inherent limitations hinder the possibility for dynamic flexible applications such as wearable medical device, activity-motion tracking sensor, electronic skin (E-skin), interactive controller, and self-powered sensor system [63]. Incorporating micro- or nanoscale ceramic particles into polymer matrices serves as an effective strategy to retain the desirable electrical characteristics of ceramics while imparting enhanced mechanical flexibility from the polymer phase to overcome the intrinsic brittleness and rigidity of conventional ceramic-based TENG devices.

Ceramic materials offer unique advantages in TENG applications through their high dielectric constants, mechanical robustness, and ability to maintain stable surface properties under harsh operating conditions [64]. Chemical engineering plays a pivotal role in controllably integrating these ceramics as functional fillers into polymer matrices to create high-performance composites. Jia et al. reported that integrating amino-functionalized $BaTiO_3$-NH_2 nanoparticles with two-dimensional graphene in polydimethylsiloxane (PDMS) matrices improved surface morphology and enhanced electrical output characteristics of TENGs (Fig. 4d–e) [65]. Scanning electron microscopy confirmed uniform anchoring of $BaTiO_3$-NH_2 nanoparticles onto graphene lamellae, mitigating agglomeration typically observed in ceramic filler systems. This homogeneous dispersion, achieved through advanced dispersion and interfacial engineering techniques, produced finely textured and porous surfaces in PDMS/graphene/$BaTiO_3$-NH_2 hybrids, enlarging real contact interfaces and augmenting charge induction and separation at triboelectric junctions. The high permittivity of $BaTiO_3$-NH_2 increased the dielectric constant of composites from 2.75 to 5.52 (Fig. 4f). This increase improved charge storage density and enhanced open-circuit voltage and current output of devices. Furthermore, the PDMS/graphene/$BaTiO_3$–NH_2 composite film exhibited a strength modulus of 0.84 MPa and a maximum strain of 150 %, which are 3-fold and 1.25-fold higher than those of the PDMS/$BaTiO_3$ film. The chemical functionalization with amino groups, a targeted surface modification strategy, promoted charge trapping and reinforced dipolar polarization at interfacial regions, supporting superior electrical stability and durability under repeated operation.

Venkatesan et al. integrated nickel-doped cobalt ferrite (Ni/$CoFe_2O_4$, N-CF) ceramic nanoparticles into electrospun PVDF nanofibers to enhance triboelectric performance (Fig. 4g–h) [66]. Scanning electron microscopy images confirmed that both CF and N-CF nanoparticles exhibited uniform spherical morphologies with diameters in the tens of nanometer range. Through careful formulation engineering, the increased conductivity of electrospinning solutions due to nanoparticle incorporation directly contributed to fiber diameter reduction. Furthermore, strong interfacial ion-dipole interactions between ceramic nanoparticles and PVDF matrices induced β-phase formation and enhanced charge-trapping capabilities. Electrical measurements showed that pristine PVDF/Al TENG devices had lower performance (V_{OC} = 22 V, I_{SC} = 0.61 μA) compared to the Al/PVDF/N-CF (Al/PNC3) TENG devices (V_{OC} = 421 V, I_{SC} = 1.0 μA). Despite 25,000 cycles, the output performance of the Al/PNC3-TENG device remained constant. The synergistic effects of Ni-doped CF fillers improved charge-trapping efficiency and reduced triboelectric losses.

Khosroshahi et al. demonstrated effective TENG performance enhancement through surface morphology modification of melamine foam-based nanocomposites via aluminum oxide (Al_2O_3) nanoparticle incorporation (Fig. 4i) [67]. The SM (MF-PVA-Al_2O_3) composite exhibited Brunauer-Emmett-Teller surface area of 3.94 cm^2/g, representing a two-fold increase attributed to Al_2O_3 nanoparticle presence. Furthermore, total pore volumes for MF and nanocomposites were 0.008 and 0.011 cm^3/g, respectively. This TENG exhibited 1325.96 V, 7.71 μA

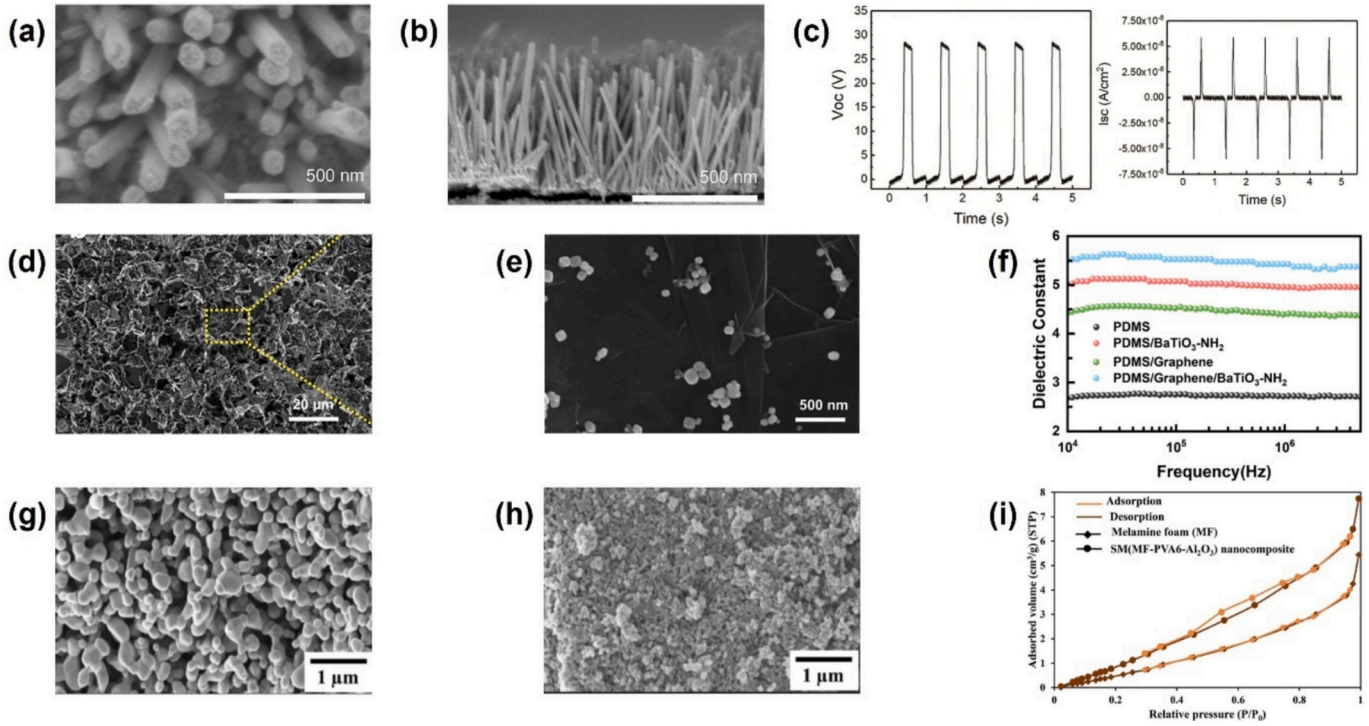


Fig. 4. (a) Top-view and (b) cross-sectional SEM image of Ga-doped ZnO NRs arrays. (c) Voltage and current density output of TENG made of PDMS-4 M% Ga-doped ZnO NRs arrays. Reproduced with permission [62]. Copyright 2019, Elsevier B.V. (d) SEM image and (e) magnified SEM image of the graphene/$BaTiO_3$-NH_2 composite. (f) Dielectric constant of various composites for TENG active layer. Reproduced with permission [65]. Copyright 2025, American Chemical Society. (g) SEM image of NiO, and (h) cobalt ferrite for enhancing interfacial ion-dipole interaction in TENG active layer. Reproduced with permission [66]. Copyright 2025, Springer Nature. (i) BET isothermal graph for showing the enhanced porosity of SM (MF-PVA-Al_2O_3) nanocomposite. Reproduced with permission [67]. Copyright 2024, American Chemical Society.

and 70.99 mW/cm^2 and was capable of powering 110 commercial LEDs. In addition, it maintains a stable output for 64,800 cycles (270 min). These results indicate that Al_2O_3 incorporation, a composite formulation strategy, does not negatively impact MF porosity while enhancing surface area, which plays crucial roles in triboelectric output enhancement. The integration of ceramic nanoparticles and surface modification strategies reveals that controlled dispersion and functionalization, both core chemical engineering challenges, are critical factors in achieving enhanced dielectric properties while maintaining favorable surface morphologies for charge generation [68].

2.3. Polymer materials

Polymer materials represent the most versatile and widely adopted category in TENG fabrication, with their exceptional potential unlocked through chemical and process engineering for mechanical flexibility, and extensive opportunities for surface morphology modification [69,70]. He et al. fabricated microstructured TENGs employing PDMS as active materials (Fig. 5a–b) [71]. Due to superior flexibility, facile processability, low cost, and stable electrical properties, PDMS represents one of the most extensively utilized polymers in TENG fabrication. Integration of surface patterning techniques significantly enhances output performance. Metal woven test sieves were employed as molds, with a modified thermal embossing process (a classic polymer processing technique) applied to generate microgrid patterns on PDMS surfaces. Compared to TENGs composed of smooth PDMS films (S-TENG), devices with microgrid-patterned PDMS (X-TENG) exhibited 1.5-fold enhancement in I_{SC} (2.72 μA) and 2.3-fold improvement in V_{OC} (62 V). Tan et al. fabricated ultralight, porous, high-performance TENGs utilizing silk fibroin (SF) aerogels and PDMS sponges as friction layers. At SF concentrations of 3 %, specific surface areas reached peak values of 8.4 m^2/g, achieving maximum open-circuit voltage, short-circuit current, and transferred charge. Increasing SF concentrations above 4 % led to specific surface area reductions, resulting in output performance degradation. The porous SF aerogel based TENG exhibited optimal output performance at 3 % SF concentration, achieving a V_{OC} of 365 V, a I_{SC} of 11.8 μA, and a P_{max} of 7.52 W/m^2. These results confirm that engineering the material's specific surface area, a common goal in chemical engineering for applications like catalysis and adsorption, directly correlate with SF aerogel TENG performance enhancement (Fig. 5c) [72].

Lei et al. reported that surface morphology of PA66 and ethyl cellulose (EC) nanofibers had pronounced influences on output characteristics of nano/microporous fiber-based TENGs. Pure PA66 nanofibers possessed smooth surfaces with appreciable diameters and uniform distributions (Fig. 5d) [73]. Mean fiber diameters were only 110 nm, demonstrating excellent electrospinning performance. EC nanofibers exhibited rough fiber surfaces with uneven diameter distributions (Fig. 5e). At PA66/EC ratios of 5:5, PA66/EC fibers were smooth with a mean diameter of 303 nm and had good flexibility (Fig. 5f). The optimal PA66/EC ratio produced a V_{OC} of 866 V and an I_{SC} of 3.5 μA which were 5.8 and 3.3 times those of the pure PA66 film. Such nano-structural control, achieved by optimizing chemical formulations and process parameters, improved specific surface areas and induced higher surface charge densities. Mohamadbeigi et al. fabricated nanofiber structures employing polyethylene oxide (PEO) as polymer-based materials (Fig. 5g) [74]. PEO/Cu_2O composite nanofibers (PCNFs) exhibited uniform fibrous networks with average diameters of approximately 266 nm, providing high surface areas and continuous morphologies. The electric current of the PCNF increased significantly from 0.4 μA to 13 μA as the relative humidity (RH) value from 30 % to 99 %. The continuity and uniformity of as-spun nanofibers resulted in structures with high aspect ratios and large specific surface areas, essential characteristics for achieving sensitive breath sensor applications. Le et al. reported TENG

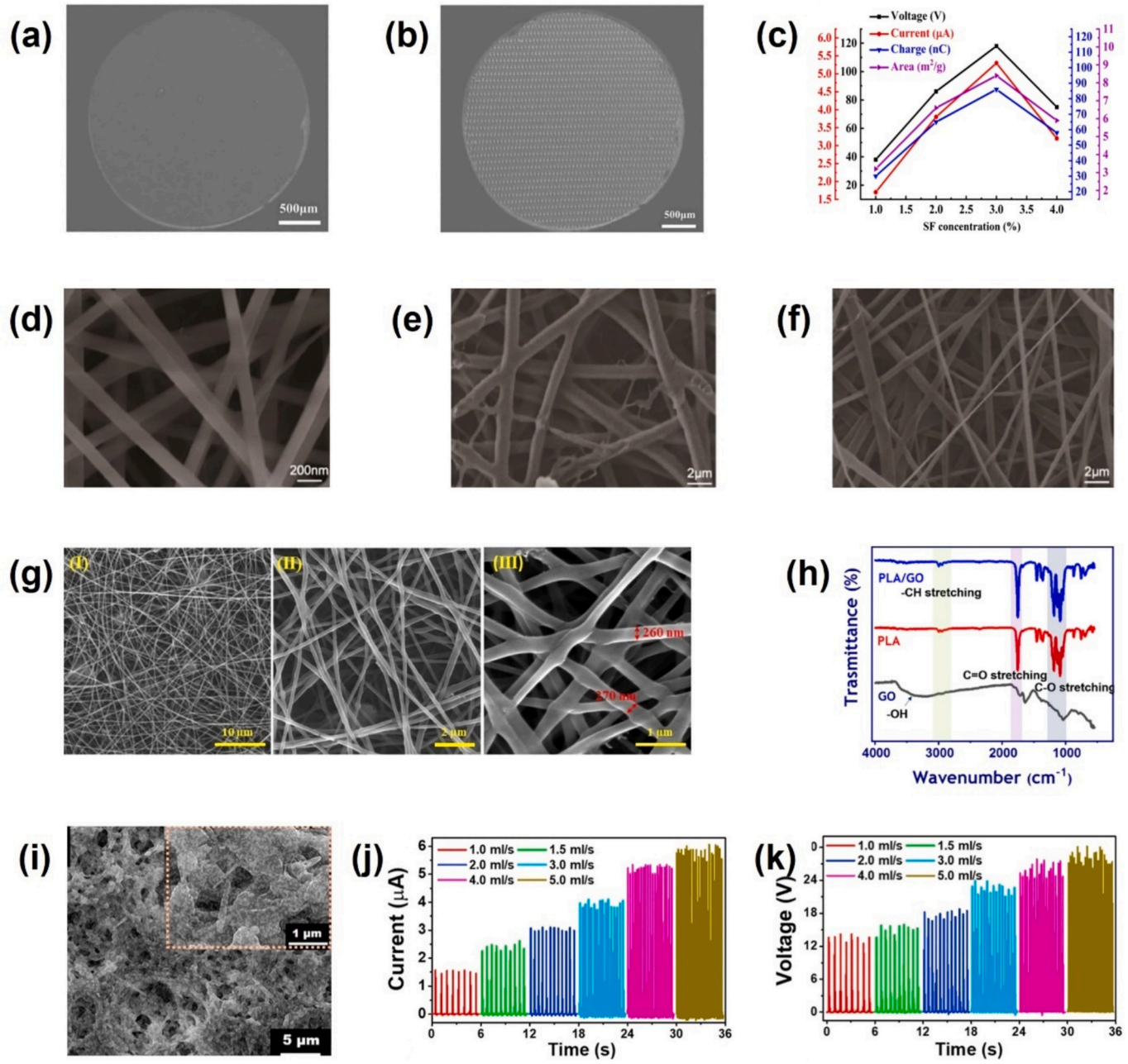


Fig. 5. (a) SEM image of flat PDMS film, and (b) microgrid PDMS for TENG active layer. Reproduced with permission [71]. Copyright 2023, Elsevier B.V. (c) Variation trends of voltage, current, charge, and specific surface area as a function al SF concentrations in TENG active layer. Reproduced with permission [72]. Copyright 2025, American Chemical Society. SEM image of (d) PA66 nanofiber, (e) EC nanofiber, and (f) PA66/EC nanofiber for TENG material. Reproduced with permission [73]. Copyright 2025, American Chemical Society. (g) SEM images of PCNFs with various magnifications. Reproduced with permission [74]. Copyright 2025, Springer Nature. (h) FTIR spectra of PLA, GO and GO/PLA nanocomposite, which is used to increase surface area. Reproduced with permission [75]. Copyright 2023, Elsevier B.V. (i) FE-SEM image of FSiP membrane surface, (j) current (k) and voltage output of the FSiP-TENG with the various rate of water flow. Reproduced with permission [76]. Copyright 2021, Elsevier B.V.

using a surfactant-free graphene oxide-polylactic acid (GO/PLA) nanocomposite. GO nanoparticles were dispersed into DMF and subsequently mixing with PLA to fabricate a honeycomb-GO/PLA film. FTIR spectra confirmed characteristic bands of GO (C═O, C—O, –OH) were observed together with the absorption peaks of PLA, comparative analysis demonstrating TENG based on honeycomb-GO/PLA achieved a maximum output power of 3.25 mW, about 13.6 times higher than the flat-surface device (Fig. 5h) [75]. Furthermore, it demonstrated excellent humidity resistance, maintaining high output performance at an 80 % humidity. These results indicate that PLA, when formulated as a composite through chemical engineering methods, effectively enhances TENG performance.

The ferroelectric polymers have been widely employed in conventional TENGs with the excellent electrical properties including high dielectric constant, intrinsic polarization and mechanical flexibility. PVDF is considered as a TENG with the high negative triboelectric properties, mechanical robustness, and chemical stability. Vu et al. demonstrated liquid-solid TENG surface polarity tuning through epitaxial growth on PVDF membrane as described at Fig. 4(i). The two-step surface modification was conducted to surface morphological engineering with the combination of Silica nanoparticles (SiNPs) coating with epitaxial growth method and 1H,1H,2H,2H-

Perfluorooctyltriethoxysilane (FOTS) implantation to achieve the durability, stability and water-resistance. The modified FOTS/SiNP TENG(FSiP-TENG) delivers outstanding triboelectric output, generating 5.79 μA of current and 28.3 V of voltage at a flow rate of 5.0 mL/s. Notably, the peak power density reaches 420 mW/m^2, which is 2.7 times and 10.8 times greater than that of the pristine PVDF-TENG (Fig. 5j–k). The results indicate the modification availability of PVDF-based TENG with the field of application such as water flowrate monitoring, ocean and wave energy harvesting, and water-resistant operation [76].

The comprehensive analysis of metallic, ceramic, and polymer materials demonstrates that surface morphology engineering transcends simple material selection, requiring careful application of chemical engineering principles for interfacial properties and structural design. Metallic materials excel in providing conductive pathways and can achieve remarkable performance through nanostructuring, while ceramic materials offer unique advantages in dielectric enhancement and mechanical stability. Polymer materials provide unparalleled design flexibility and processability, enabling complex surface architectures that maximize charge generation. The synergistic effects observed in hybrid material systems highlight the potential for further performance optimization through intelligent material combination and surface engineering strategies. The diverse polymer systems examined illustrate that surface area enhancement through nanostructuring, whether achieved via polymer processing like electrospinning, patterning, or composite formation, consistently correlates with improved triboelectric performance across different polymer types.

3. Processes

The realization of optimized surface morphologies requires precise control over fabrication processes, which directly influence the final structural characteristics and performance of TENG devices [77]. From a chemical engineering standpoint, these processes are considered unit operations that can be designed, controlled, and scaled to achieve specific material outcomes. Surface morphology engineering encompasses a diverse range of processing methodologies, each offering distinct advantages and limitations depending on the target application and material system. The selection and optimization of these methods are central to process engineering, aiming for reproducibility, scalability, and cost-effectiveness. This section categorizes and analyses the primary processing approaches (e.g., physical, chemical, and hybrid methods) that enable controlled modification of triboelectric interfaces, examining their underlying chemical engineering mechanisms, capabilities, and impact on device performance.

3.1. Chemical processing

Chemical processing represents a bottom-up approach in surface morphology engineering, leveraging molecular-level transformations to fundamentally alter material properties. These methods, rooted in reaction engineering, polymer science, and formulation chemistry, allow for precise tuning of surface chemistry, hydrophobicity, and charge-trapping characteristics that complement physical structural modifications. Unlike physical methods that reshape existing material, chemical processes create new functionalities, offering a powerful toolkit for advanced material design [78,79]. Yang et al. reported TENGs (CNT-TENGs) composed of thermoplastic polyurethane (TPU) films doped with carbon nanotubes (CNTs) and polytetrafluoroethylene (PTFE) as friction materials. This work is a prime example of chemical engineering in composite formulation and dispersion technology. TPU particles were initially dissolved in *N*,*N*-dimethylformamide (DMF) solvent and completely dissolved after 36 h. The choice of DMF is critical; its solubility parameters are carefully selected to effectively solvate the long-chain TPU polymer, allowing it to uncoil and create a homogeneous solution, a fundamental concept in polymer solution thermodynamics. CNTs were mixed into DMF, with CNT-infused DMF and DMF-TPU solutions mechanically stirred, followed by 30 min of ultrasonication to ensure uniform CNT dispersion within TPU solutions. These mixing and sonication steps are crucial mass transfer operations designed to overcome the strong van der Waals forces that cause CNTs to agglomerate. According to AFM analysis, the initial roughness of the TPU and PTFE surfaces was 229.2 nm and 172.6 nm, respectively. Even after 100,000 cycles, these values remained approximately 199.9 nm and 179.9 nm, respectively. Notably, surface morphology changes were minimal even after long-term operation, suggesting superior durability during use. Scanning electron microscopy images revealed CNT microstructures. CNT-TENGs achieved optimal electrical performance (V_{OC} = 146 V, I_{SC} = 0.21 μA, and P_{max} = 13.5 mW/m^2) with CNTs integrated into TPU matrices at 0.75 % mass fractions (Fig. 6a–c) [80]. This specific concentration represents an optimized process parameter, balancing the need for a conductive CNT network against the risk of degrading the polymer matrix's mechanical integrity at higher loadings, a typical trade-off analysis in materials process engineering.

Nie et al. reported excellent triboelectric charge density and hydrophobicity achieved on cellulose nanofibrils (CNFs) through simple and environmentally friendly aminosilane modification approaches for CNF films. This study showcases the power of surface reaction engineering to impart desired properties. Incorporating amino groups into CNF improved surface polarizability of cellulose-based materials. The first step involved silanol release with water to form reactive hydroxyl groups, a hydrolysis reaction whose rate is dependent on pH and temperature. The second step minimized condensation to ensure maximum hydroxyl active sites, requiring careful control over reaction kinetics to favor surface grafting over premature self-polymerization of the silane in solution. The third step involved polysiloxane structure formation through silanol group self-condensation, making CNF surfaces 3-(2-aminoethylamino)propyldimethoxymethylsilane (AEAPDMS) modified. After modification, the surface structure was transformed into a sheet-like structure, leading to a more compact structure. Water contact angles of CNF films showed surface transitions from hydrophilic (52.1°) to hydrophobic (106.8°) after 10 % AEAPDMS modification, improving hydrophobicity of AEAPDMS-modified CNF films (Fig. 6d–e) [81]. This significant change in surface energy is a direct quantitative measure of the success of the molecular-level surface functionalization.

Chemical grafting modifies the surface chemistry of triboelectric materials by covalently attaching functional groups or side chains, thereby tuning their electron-donating/accepting characteristics and enhancing charge transfer during contact electrification. Byun et al. demonstrated that even a monolayer of less than 1 nm thickness functionalized with electron-donating (–NH_2, –SH), neutral (–CH_3), or electron-withdrawing (–CF_3) groups could significantly shift both the amount and polarity of triboelectric charge in SiO_2, effectively altering the conventional triboelectric series (Fig. 6f) [82]. Kelvin probe force microscopy (KPFM) analysis revealed direct correlation between the induced surface dipole moments and the changes in triboelectric output, confirming that surface potential and electronic state are key governing factors in charge transfer during contact electrification (Fig. 6g–h). This surface modification strategy greatly expands the design space for TENGs, allowing for selection and engineering of materials with targeted triboelectric polarities, thereby enhancing device flexibility and output performance.

Sol-gel processing is a versatile chemical method used to fabricate thin films, hybrid coatings, and nanostructured layers. This process has the advantages such as excellent film uniformity, cost-effectiveness, scalability, and low process temperature. In the aspect of surface morphological engineering, sol-gel processing improves the output performance of the TENG with the hierarchical and nanostructured design, composite layer and chemical modification. Lee et al. presented hierarchically surface-textured film with rapid hydrolytic photoreaction. The surface-textured glass fabric reinforced siloxane hybrid (SGH) film in large scale was fabricated with sol-gel reaction of methoxy groups of 3-(2-Trimethoxysilylethyl)cyclohexene oxide (ECTS) and

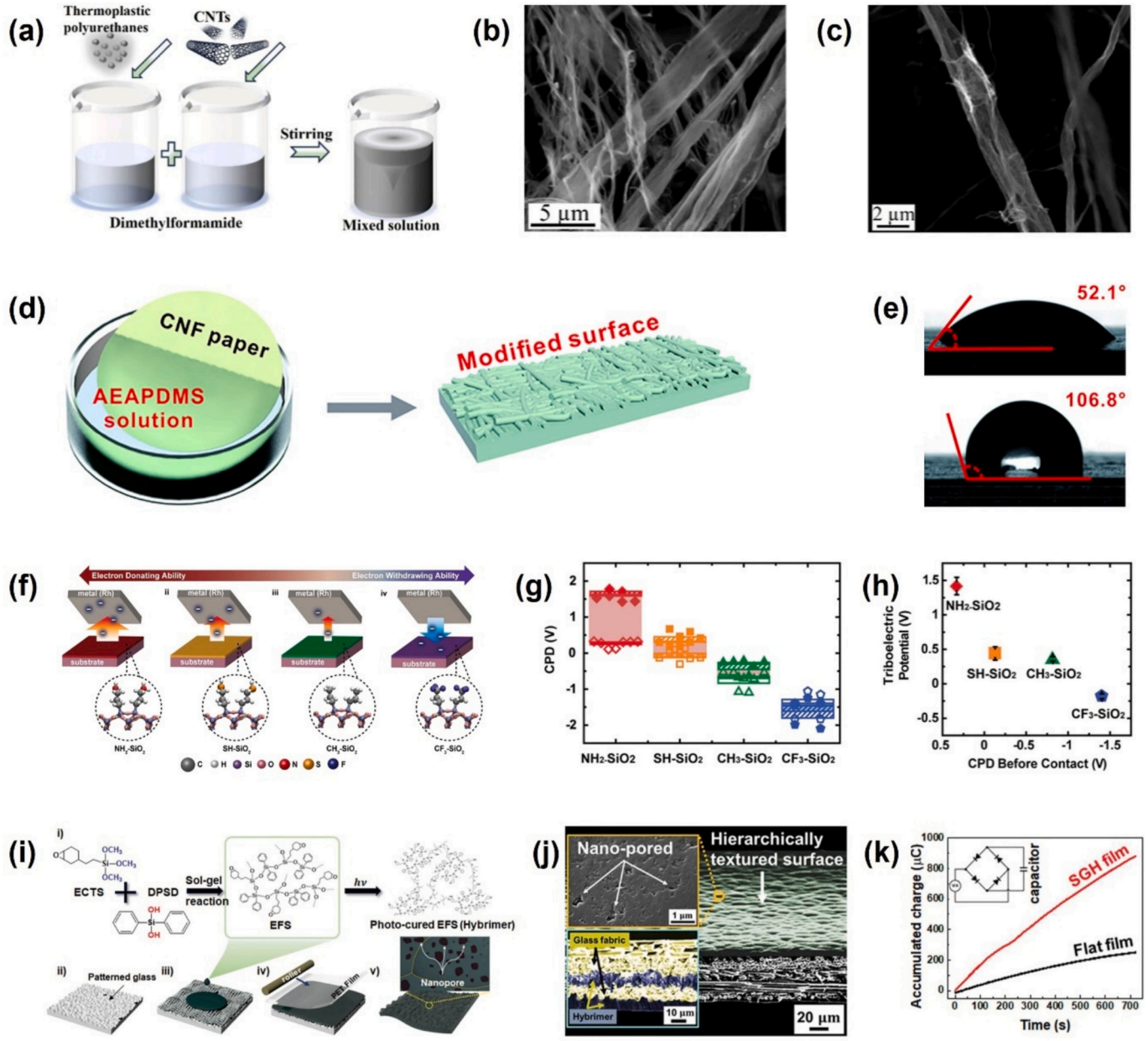


Fig. 6. (a) Fabrication process of CNT-doped TPU nanocomposite. SEM image of CNTs for nanocomposites with at (b) 10,000×, and (c) 15,000× magnification. Reproduced with permission [80]. Copyright 2025, American Chemical Society. (d) Schematic illustration of AEAPDMS-modified of CNF paper and its magnified surface. Water contact angle of (e) raw CNF and of AEAPDMS-modified CNF. Reproduced with permission [81]. Copyright 2020, American Chemical Society. (f) Schematic of triboelectrification on SAM-modified substrates. (g) Contact potential difference (CPD) result analyzed with KPFM. (h) Triboelectric potentials of the surfaces before contact. Reproduced with permission [82]. Copyright 2016, American Chemical Society. (i) Schematic illustration of SGH film fabrication. (j) SEM image of hierarchical microdome-nanopore structure. (k) Charged capacitance comparison with SGH film and bare film. Reproduced with permission [83]. Copyright 2020, John Wiley and Sons.

silanol groups of dihydroxydiphenylsilane (DPSD) to synthesis epoxy-functionalized siloxane (EFS) resin as illustrated in Fig. 6(i). The pressurized EFS on mold was photo-cured by UV light and created microdome and random nanopores as hierarchical structure (Fig. 6j). The nanogenerator with SGH film was tested by recharging the commercial capacitor, and successfully charged the capacitor within ≈700 s up to 880 μC, which was about 4 times faster than the hybrimer-based flat TENG as shown in Fig. 6(k) [83].

Chemical modification strategies, as executed through the precise control of reaction conditions and formulation that are central to chemical engineering, prove essential for optimizing surface properties beyond structural considerations, with functionalization and doping approaches providing pathways to enhanced charge stability and environmental resistance [84,85].

3.2. Physical processing

Physical processing methods provide direct mechanical or energy-based approaches to surface modification, representing a top-down strategy for creating micro- and nano-scale features. These methods offer precise control over structural features without altering the fundamental chemical composition of materials. Many of these techniques, however, are governed by chemical engineering principles such as fluid dynamics, heat transfer, and rheology [86]. Wajahat et al. reported laser-based physical surface patterning strategies to enhance triboelectric performance of three-dimensional printed nanogenerators.

Polyamide 12 (PA12) and VeroClear substrates were patterned into line, circular, and pyramid geometries using precise laser irradiation. Well-defined and uniform surface morphologies were obtained, with pyramid geometries exhibiting optimal performance. Pyramid-patterned devices demonstrated V_{OC} of 95.8 V and I_{SC} of 0.99 μA, corresponding to approximately 26 % and 13.7 % improvements over non-patterned counterparts (Fig. 7a–c) [87].

Memon et al. employed physical processes including electrohydrodynamic (EHD) printing for ZIF-67-based TENG fabrication (Fig. 7d–e) [35]. EHD printing, a non-contact deposition method, utilizes high electric fields to atomize inks and form uniform thin films without requiring thermal treatment or chemical reactions. The success of this technique relies heavily on the chemical engineering of the ink, including its viscosity, surface tension, and conductivity, which dictate the electrohydrodynamic behavior. Surface morphology of aluminum substrates before and after EHD deposition revealed uniform and densely coated ZIF-67 distributions. ZIF-67 particles have well-defined plate-like structures with sharp edges and corners. The fabricated ZIF-67 based TENG exhibited a V_{OC} of 119.6 V and an I_{SC} of 7.7 μA under optimized conditions. Such physical processes facilitate effective contact area increases, contributing to charge generation efficiency enhancement at triboelectric interfaces.

Wu et al. presented multi-mode stretchable and wearable triboelectric nanogenerators (msw-TENGs) for biomechanical energy harvesting and physiological function sensing. The fabrication process involved three-dimensional printing molds, silicone perfusion, silicone film peeling, and liquid metal injection (Fig. 7f) [88]. Molds were fabricated by three-dimensional printing, with Ecoflex 00-10 AB adhesive mixed in 1:1 ratios. Mixtures were vacuum-treated to remove bubbles and injected into molds, then placed in ovens at 30 °C for 2 h to obtain silicone films. Two different silicone film structures were assembled using adhesive mixtures. This device is designed with a sandwich-type island-bridge structure, providing mechanical flexibility and elasticity. The msw-TENGs achieved power outputs of 15 μW in contact-separation mode. Moreover, the device maintained stable output after 10,000 cycles at 1 Hz.

Kim et al. fabricated polyimide (PI) nanofibers via electrospinning and utilized them as friction layers for enhanced TENG performance. PI nanofibers were electrospun from PI inks prepared from polyimide powder. No fibrous structures formed at 5 wt%, indicating electrospraying occurred (Fig. 7g). Uniform nanofibrous structures formed at 15 wt% and 20 wt% concentrations (Fig. 7h–i) [89]. This transition is a classic example of process control in chemical engineering, where solution rheology and polymer chain entanglement, governed by concentration, are manipulated to control the final morphology. When electrospun PI nanofibers were employed, voltage increased by approximately 5.5-fold and current enhanced by 3.9-fold compared to film-like structures. Physical processing techniques demonstrate excellent reproducibility and scalability potential, with laser patterning and electrospinning emerging as particularly effective methods for achieving controlled surface morphologies with significant performance improvements [90,91].

3.3. Chemical and physical hybrid processing

The combination of physical and chemical processing methods offers

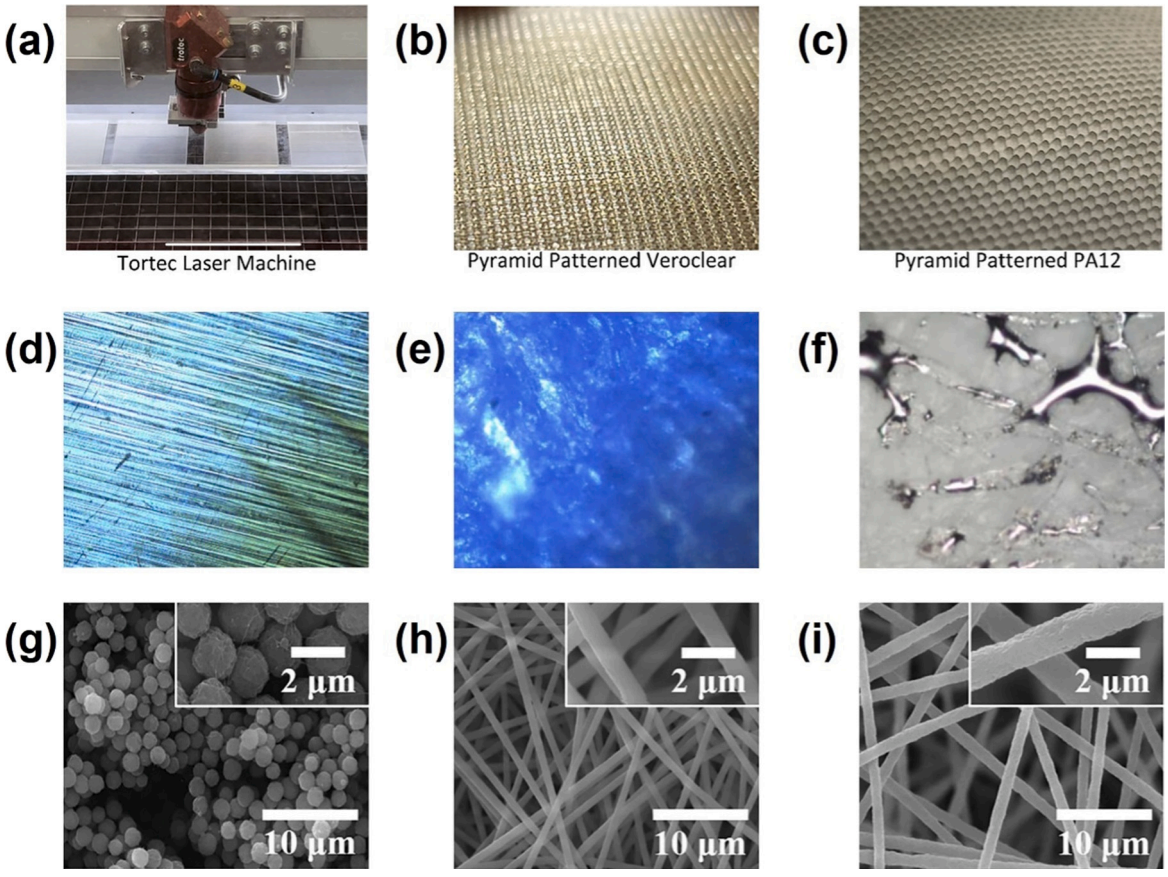


Fig. 7. (a) Tortec laser machine for patterning line, circular, and pyramid patterns on TENG materials. Optical image of (b) pyramid patterned Veroclear, and (c) PA12. Reproduced with permission [87]. Copyright 2024, John Wiley and Sons. Aluminum substrate (d) before and (e) after EHD process and ZIF-67 coating. Reproduced with permission [35]. Copyright 2024, American Chemical Society. (f) Liquid metal microstructure for multi-mode stretchable and wearable triboelectric nanogenerators. Reproduced with permission [88]. Copyright 2021, Elsevier B.V. SEM image of electrospun nanofiber membranes with PI concentration of (g) 5 wt %, (h) 10 wt%, and (i) 20 wt%. Reproduced with permission [89]. Copyright 2024, Springer Nature.

synergistic advantages that exceed the capabilities of individual approaches, enabling simultaneous optimization of structural and chemical surface properties [92]. This strategy, akin to process integration in chemical engineering, combines the top-down architectural control of physical methods with the bottom-up molecular functionalization of chemical methods.

Meena et al. demonstrated that coupled transfer printing-spin coating techniques could imprint wrinkled silicone structures with tunable periodicity and amplitude directly onto PVDF, enhancing tactile sensitivity. In physical processes, spin coating highlighted imprinted PVDF formation. In chemical processes, H_2 plasma and UV/ozone treatments were applied to PDMS surfaces, while 99.5 % *N*,*N*-dimethylformamide (DMF) was utilized as PVDF solvent. This hybrid approach leverages physical templating to create the macro-structure and then uses chemical surface activation (plasma) to enhance interfacial properties. The device featured a PVDF thin film with micro-periodic wrinkle structures transferred onto its surface. The wrinkled PVDF thin film exhibited strong piezoelectricity arising from its β-phase crystallinity and enhanced triboelectric activity resulting from the increased surface area and localized electric fields, thereby enabling efficient hybrid charge generation. These processes achieved hybridized electrical outputs approximately 200 % higher than pristine PVDF substrates (Fig. 8a–b) [93].

Jakmuangpak et al. reported bio-TENG fabrication based on bacterial cellulose (BC) nanocomposites through gradual drying of BC hydrogels on conductive substrates (Fig. 8c–d) [94]. In physical processes, BC films were adhesively bonded on indium tin oxide (ITO) substrates directly without conductive pastes and dried in ovens at 60 °C. This drying step is a carefully controlled heat and mass transfer operation. In chemical processes, simple drying of BC/ZnO (ZBC) nanocomposite hydrogels could not produce sufficient adhesion between ZBC films and ITO substrates. To address this issue, a chemical engineering solution was applied: ZBC hydrogel surfaces were modified with amino-silane to create covalent bonds at the interface, solving an adhesion problem that the physical process alone could not. The ZBC film structure exhibited increased surface roughness and porosity due to the dispersion of ZnO nanoparticles within the BC nanofiber network. Bio-TENGs based on BC and ZBC films generated high V_{OC} (57.6 V) and I_{SC} (5.78 μA) compared to BC-TENGs fabricated through other routes.

Electrodeposition was considered as the basic method for surface morphological engineering since the initial stage of TENG. The method enables accurate tuning of surface structure and composition in triboelectric layers with boosting charge density and ensuring consistent output. Additionally, this technique provides a scalable, low-temperature, and economical manufacturing method accommodates complex shapes and flexible substrates, enhancing device integration and durability. Zhu et al. demonstrated a low-cost TENG with coupled process of contact charging and electrostatic induction by pulse electrodeposition (PED) of silver as depicted in Fig. 8(e). The micro/nano-crystalline silver was deposited with PED on Kapton film to achieve uniform coating quality, densely packed structure, even with the property regulation with the pulse control. This process accomplished high output performance with the power density of 31.2 mW/cm^3 and maximum open circuit voltage of 110 V (Fig. 8f–g) [95].

The examination of processing methodologies reveals that optimal TENG performance often requires careful integration of multiple processing approaches rather than reliance on single techniques. Physical processes provide precise structural control and excellent reproducibility, making them suitable for large-scale manufacturing applications. Chemical processes enable molecular-level modifications that enhance charge trapping and surface stability, while hybrid approaches embody a core principle of chemical process design: combining unit operations synergistically to surpass the limitations of individual methods. The selection of appropriate processing strategies must consider not only

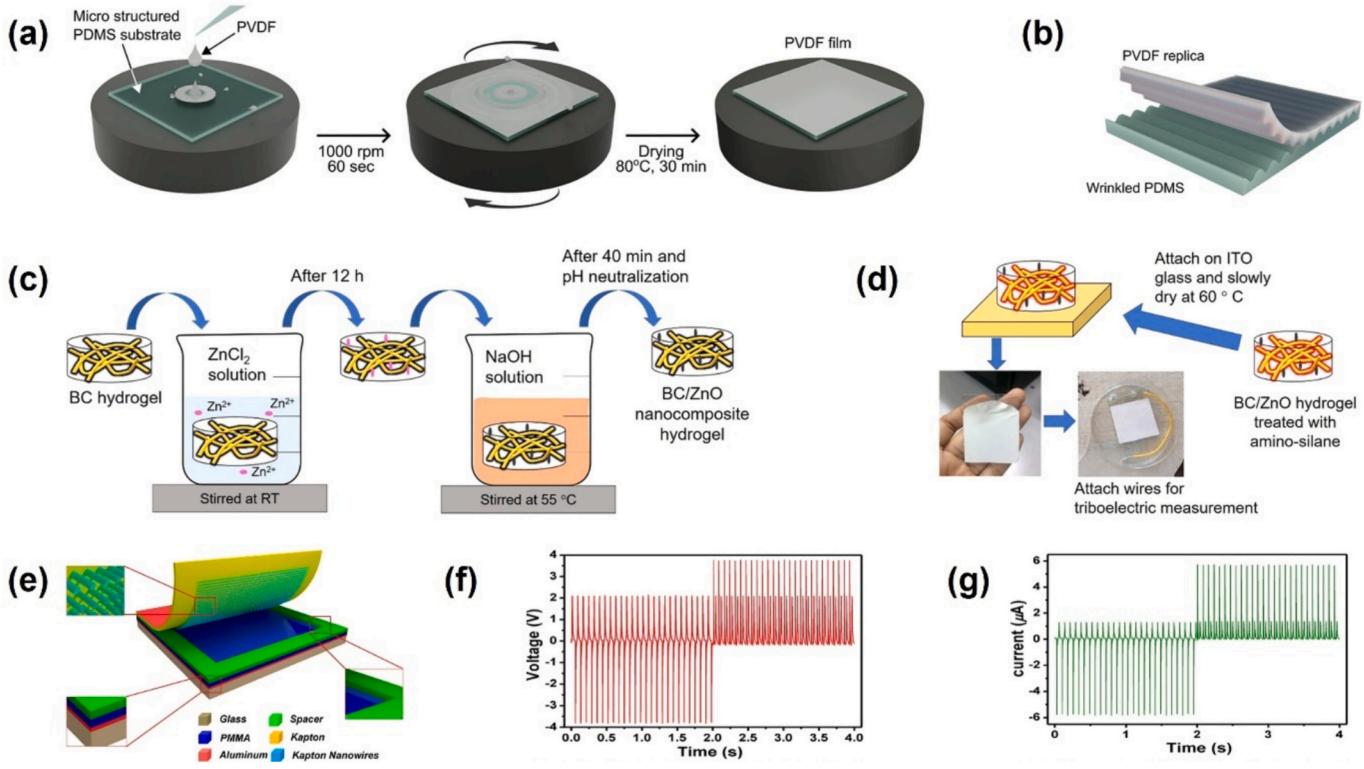


Fig. 8. (a) Schematic procedure of a microstructured PVDF films, and (b) a wrinkled PVDF structure. Reproduced with permission [93]. Copyright 2024, John Wiley and Sons. (c) Fabrication process of an amino-silane-treated BC/ZnO (ZBC) nanocomposite hydrogel, and (d) its attachment onto the ITO glass for TENG measurement. Reproduced with permission [94]. Copyright 2020, American Chemical Society. (e) Generator design layout and materials specification. (f) Voltage output performance and (g) Current output performance of TENG. Reproduced with permission [95]. Copyright 2012, American Chemical Society.

performance targets but also manufacturability, cost-effectiveness, and long-term stability requirements for successful commercialization. Hybrid processing demonstrates the potential for achieving superior performance through carefully designed multi-step procedures that leverage the complementary strengths of different modification techniques.

4. Structural engineering

The architectural design of triboelectric surfaces represents a critical dimension in TENG optimization, where the precise control of structural features across multiple length scales determines the effectiveness of charge generation and transfer processes. From a chemical engineering viewpoint, this is a multiscale design challenge, analogous to designing a chemical reactor where performance is dictated by phenomena occurring from the molecular level to the macroscopic level. Surface morphology engineering through structural design encompasses micro-, nano-, and hierarchical architectures, each contributing distinct advantages to triboelectric performance enhancement [96–98]. While theoretical TENG models assume an ideal configuration consisting of two rigid, smooth surfaces with extensive contact area, practical applications present significantly different interfacial conditions. In realistic TENG devices, contact surfaces deviate substantially from the idealized flat geometry and exhibit pronounced deformation under applied pressure, making surface roughness a critical parameter governing electrical output performance. The triboelectric layers commonly employed in practical devices, particularly polymers such as PDMS and nylon, possess relatively low elastic moduli that enable significant surface deformation under operational pressures. Consequently, the apparent contact area differs markedly from the real contact area, with this discrepancy becoming increasingly pronounced as surface roughness increases [51]. This section systematically explores how different structural approaches, many of which are realized through carefully designed chemical and physical processes, influence contact mechanics, charge density distribution, and overall device efficiency, providing insights into optimal design principles for various applications.

4.1. Microscale structures

Microscale surface structures primarily enhance triboelectric performance by increasing effective contact areas and modifying contact mechanics during the triboelectric process [99]. These structures are often fabricated using scalable process engineering techniques that are both cost-effective and highly reproducible. Somkuwar and Kumar et al. reported lightweight, comfortable, and robust wearable TENG (W-TENG) devices fabricated using woven and knitted fabrics for self-powered technology applications. Rib-knitted structures exhibited surfaces with pronounced hills and valleys and higher roughness (S_a = 165.7 ± 10 μm, S_q = 232 ± 11 μm) compared to woven fabrics. Three-dimensional configurations of knitted loops created higher roughness than woven fabric repeat units and provided line contact instead of point contact during contact-separation processes. The knitted structure exhibited a 240.55 % increase in voltage, a 64.96 % increase in current, and a 460 % increase in power density compared to the woven structure (Fig. 9a) [100]. Wu et al. demonstrated PTFE film surface morphology modification through emery paper polishing, constructing microstructures on electrode surfaces. Contact angles between PTFE films and deionized water increased from 108 to 132° after emery paper

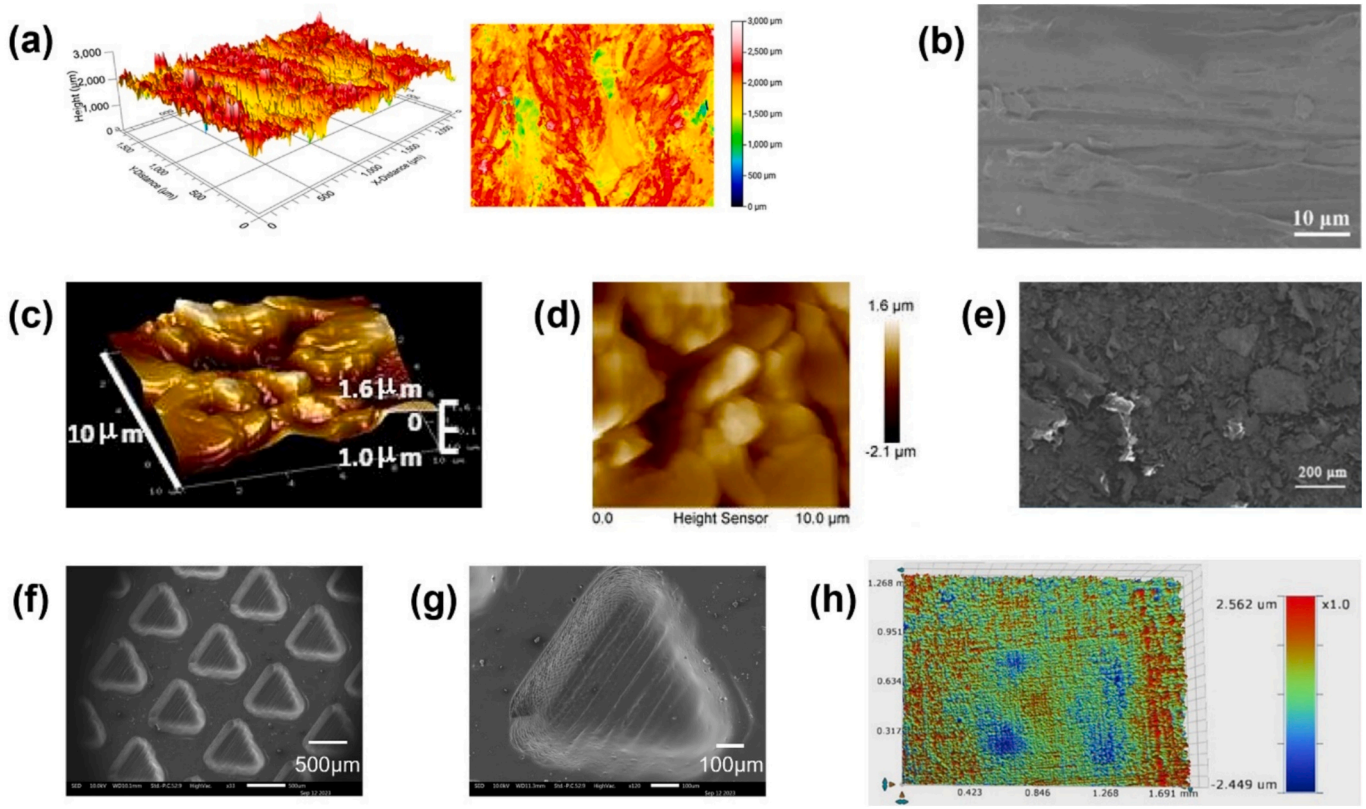


Fig. 9. (a) 3D surface profiles of a knitted nylon. Reproduced with permission [100]. Copyright 2020, American Chemical Society. (b) SEM image of the PTFE film, polished with emery papers. Reproduced with permission [101]. Copyright 2022, Springer Nature. AFM images of the electrodes, patterned by 10,000-grit sandpaper: (c) 2D, and (d) 3D image. Reproduced with permission [102]. Copyright 2017, American Chemical Society. (e) SEM images of fingernail powder surface for TENG active layer. Reproduced with permission [103]. Copyright 2020, Elsevier B.V. (f-g) SEM surface image of pyramid-patterned VeroClear (500 μm). Reproduced with permission [79]. Copyright 2024, John Wiley and Sons. (h) 3D morphology of a liquid-solid TENG. Reproduced with permission [104]. Copyright 2024, American Chemical Society.

treatment, with no residual water adsorption on surfaces. The PTFE-based TENG remained stable after 400,000 cycles of continuous operation. Both I_{SC} and V_{OC} of emery paper-treated PTFE films exceeded untreated films, indicating that output performance increased after roughening (Fig. 9b) [101]. Zhang et al. demonstrated simple, cost-effective yet efficient methods wherein friction pair materials, copper electrodes and PDMS, were both patterned using sandpaper templates (Fig. 9c–d) [102]. This approach can be viewed as a form of soft lithography or template molding, a foundational process engineering technique for replicating microscale topographies over large areas. PDMS and copper electrodes were patterned with sandpaper-like morphologies, featuring convex and concave microstructures on surfaces. The electrical output was improved several times from approximately 35 V and 15 nC to 200 V and 76 nC compared to the nonpatterned. TENGs with two-sided patterned friction layers exhibited superior output properties, suggesting that patterning microstructure morphologies improves TENG performance.

Fu et al. reported the first bio-based TENGs using human fingernails. Scanning electron microscopy images showed fingernail powder after grinding, forming microstructures composed of irregular particles with diameters ranging from 10 to 300 μm. TENGs using fingernails produced maximum V_{OC} of 87.3 V and maximum I_{SC} of 3.2 μA (Fig. 9e) [103].

Wajahat et al. reported comparisons of square, rectangular, and pyramidal patterns through COMSOL simulations, confirming that pyramidal micropatterns exhibited optimal performance. This use of computational modeling to predict the optimal geometry before fabrication is a classic example of the process design and optimization workflow central to chemical engineering. SEM analysis revealed that pyramidal microstructures formed uniform and accurate patterns (Fig. 9f–g) [79]. The pyramid pattern geometry created sharp edges that caused charge accumulation at specific points. This pattern emerged as the most efficient geometry, demonstrating solid outputs of 95.8 V (V_{OC}) and 0.99 μA (I_{SC}).

Zhou et al. presented novel methods for enhancing liquid-solid TENG (LS-TENG) output by increasing liquid-solid contact areas (Fig. 9h) [104]. FEP film surfaces after sandpaper grinding treatment exhibited increases in contour arithmetic mean deviation (R_a) from 0.079 to 0.155 μm. This treatment significantly enhanced charge transfer by increasing effective surface areas by 57.06 %, with transferred charges increasing from 46 to 73 nC and current and voltage outputs also being augmented.

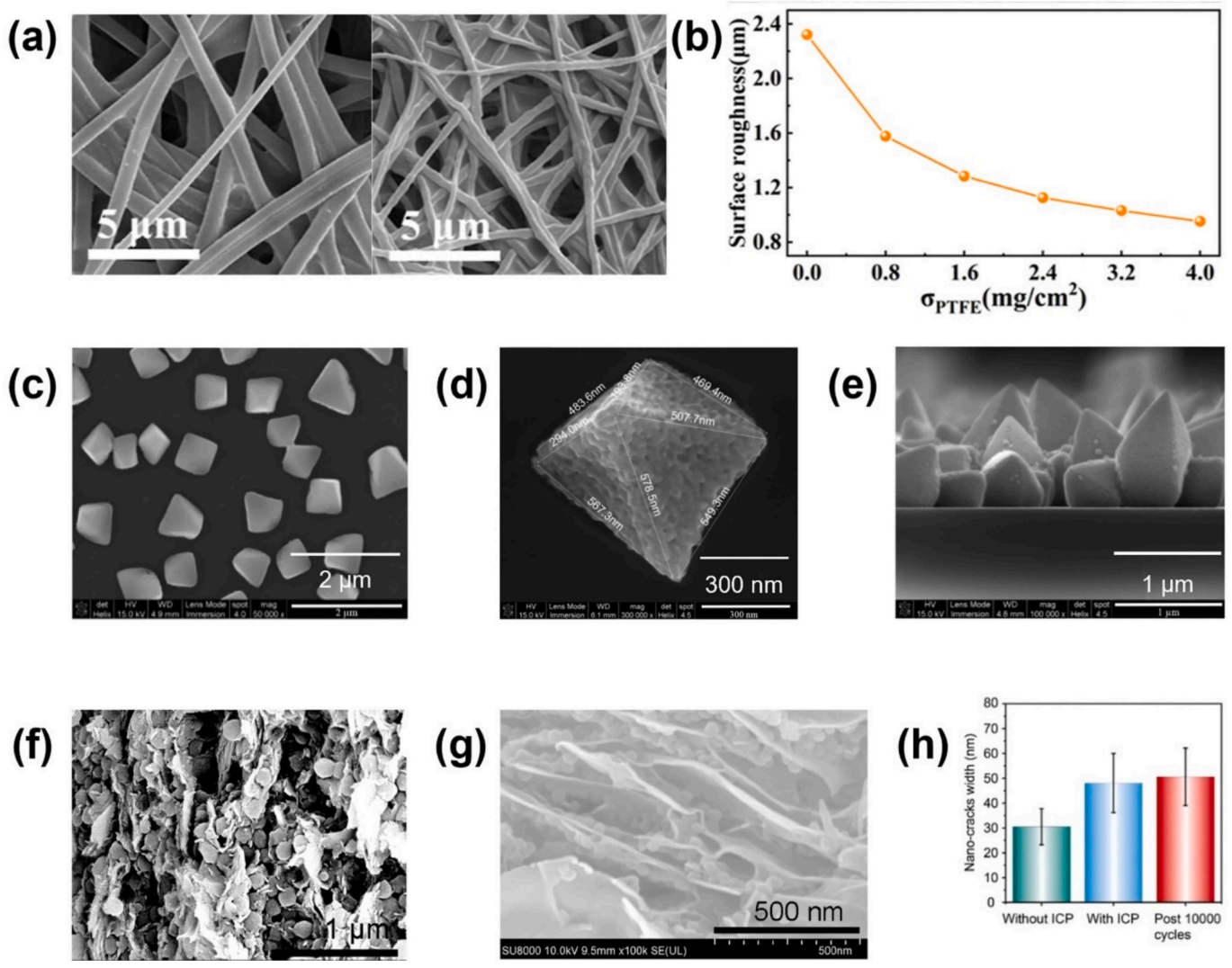


Fig. 10. (a) SEM images of EC/nylon-11 (left) and PTFE/PVDF NFs surface. (b) PTFE/PVDF NFs surface roughness. Reproduced with permission [106]. Copyright 2023, American Chemical Society. SEM images of (c) Ag octahedron nano-assemblies, (d) a magnified single Ag octahedron, and (e) Ag octahedra on ITO/PET flexible substrate. Reproduced with permission [107]. Copyright 2023, Elsevier B.V. Cross-sectional SEM image of MXene/PMMA composite film with (f) 2 %, and (g) 4 % PMMA concentrations. Reproduced with permission [108]. Copyright 2024, John Wiley and Sons. (h) Nano-cracks width on the PTFE surface of presence or absence of ICP treatment. Reproduced with permission [109]. Copyright 2022, Elsevier B.V.

These results demonstrate that increasing liquid-solid contact areas by creating microscale roughness on solid surfaces enhances LS-TENG performance. The analysis of microscale structures confirms that geometric optimization, often guided by engineering modeling and executed via scalable processing methods, provides substantial performance improvements while maintaining relatively straightforward fabrication requirements.

4.2. Nanoscale structures

Nanoscale surface features operate at length scales comparable to charge transfer distances, enabling more direct influence on charge generation, trapping, and surface potential distribution [105]. This realm of engineering represents a bottom-up design philosophy, where molecular-level control, often achieved through chemical reaction and process engineering, dictates the final material properties. Yang et al. fabricated electropositive and electronegative triboelectric layers based on ethylcellulose-coated nylon-11 (EC/nylon-11) nanofibers and polytetrafluoroethylene-coated poly(vinylidenefluoride) (PTFE/PVDF) nanofibers prepared via electrospinning and post-coating processes (Fig. 10a–b) [106]. EC/nylon-11 and PTFE/PVDF nanofibers obtained through this process had fluffy-free structures, unlike pure nanofibers. Scanning electron microscopy and three-dimensional white light optical microscopy revealed that increasing coating surface density decreased nanofiber surface roughness, resolving adhesion problems between layers. Fluffy-free nanofiber structures provided durability, mechanical flexibility, and excellent output performance. Consequently, devices achieved V_{OC} of 212 V, I_{SC} of 18.5 μA, and power density of 1.76 W/m^2, indicating electrical durability exceeding 100,000 cycles.

Navarro-Segura et al. fabricated enhanced performance TENGs using silver octahedron nano-assemblies on ITO (Fig. 10c–e) [107]. This represents a sophisticated application of controlled self-assembly, a process governed by thermodynamics and surface chemistry, which are core chemical engineering disciplines. Electrode surfaces clearly exhibited rough surfaces attributed to self-assembled silver nanoparticle presence constituting octahedral faces. Scanning electron microscopy analysis further revealed that each octahedron possessed rough surfaces formed by silver nanoparticles with average diameters of 9.5 nm. TENGs with nanostructured electrodes produced higher electrical output performance than devices with non-nanostructured electrodes. Nanostructured triboelectric generators exhibited almost 10 times higher output voltage and 65 times greater output power than TENGs with non-nanostructured contacts.

Chen et al. developed improved flexible TENGs with tile-nanostructured MXene/polymethyl methacrylate (PMMA) composite electrodes (MP-TENGs). MXene was synthesized, PMMA microspheres were prepared, then mixed to form a film, which was assembled with electrodes to fabricate the MP-TENG. The optimization of this composite is a chemical formulation engineering problem. At 4 % PMMA concentrations, devices possessed the highest output performance with output voltage of 37.8 V, output current of 1.8 μA, and transferred charge of 14.1 nC. Output performance enhanced over two-fold compared to 0 % PMMA concentrations. Enhanced output performance was attributed to charge-trapping capacity of tile nanostructures. Output performance at 20 % PMMA concentration was lower than at 0 %. Sharp decreases in output performance resulted primarily from excessive PMMA microsphere agglomeration, a common challenge in particle technology and dispersion science, which destroyed tile nanostructures and increased resistance, considerably weakening triboelectric charge generation and charge-trapping capacity. These results indicate that MXene/PMMA electrode charge-trapping capacity can be improved by precisely controlling the formulation through chemical engineering principles (Fig. 10f–g) [108].

Chen et al. reported that inductively coupled plasma (ICP) treatment on PTFE friction layers generated nano-cracks on surfaces (average width: 48.1 ± 11.9 nm), improving TENG output performance (Fig. 10h) [109]. This ICP treatment is a form of plasma reaction engineering, where chemically reactive species are generated and directed to etch the polymer surface with nanoscale precision. Smooth PTFE surfaces were roughened when treated with ICP, with nano-crack widths significantly increasing compared to untreated PTFE. Output voltages increased by approximately 2.5 times (175.7 mV vs. 50.2 mV), while signal-to-noise ratios improved from 6.4 to 12.1, representing 89.1 % increases. Nano-crack widths increased by 2.5 nm after 10,000 cycles at 7.6 % strain, revealing stability of nano-crack surface structures. Nanoscale engineering approaches demonstrate that surface roughening and nanostructure formation, achieved through advanced process control and molecular-level design rooted in chemical engineering, can achieve dramatic performance enhancements, with some configurations providing order-of-magnitude improvements in electrical output.

4.3. Hierarchical structures

Hierarchical structures combine features across multiple length scales to achieve synergistic effects that surpass the performance limitations of single-scale approaches [110,111]. This strategy epitomizes the concept of process and system integration in chemical engineering, where optimized subsystems are combined to create a superior overall system. Tao et al. reported honeycomb-inspired triboelectric nanogenerators (h-TENGs). Fabrication of nanostructured pillars on fluorinated ethylene propylene (FEP) electret thin film surfaces via ion beam etching increased contact areas during triboelectrification. Transparent polyethylene terephthalate (PET) films synthesized with conductive silver nanowires (AgNWs) were employed as flexible substrates. Through these processes, a hierarchical honeycomb-shaped arch structure with a multilayer wavy configuration was fabricated. The h-TENG exhibited excellent electrical performance with V_{OC} = 1207 V, I_{SC} = 68.5 μA, and P_{max} = 12.4 mW. High surface roughness and numerous defects between AgNWs and FEP electrets facilitated charge trapping, enhancing h-TENG performance (Fig. 11a–b) [112]. Liang et al. developed lotus leaf-inspired poly(dimethylsiloxane) bionic hierarchical structure triboelectric layers (PBTLs) fabricated using PMMA molds processed by femtosecond laser and one-step templating. This use of a natural, highly efficient structure as a biomimicry, combined with a high-fidelity templating process, is an advanced manufacturing strategy employed in chemical engineering to create complex functional materials. PBTL surfaces featured conical protrusions with approximate diameters of 24.8 μm and heights of 16 μm, covered with nanoscale particles ranging from 200 to 600 nm. PBTLs demonstrated I_{SC} of 2.15 μA, transferred charges of 15.16 nC, and V_{OC} of 48.62 V, showing 2.14-fold improvements over flat layers (Fig. 11c–d) [36]. Yin et al. fabricated hybrid-structured nanofiber membranes (HNMs) in batches using modified free-surface electrospinning (MFSE), composed of thermoplastic polyurethane/carbon black (TPU/CB) beaded nanofibers and poly(vinylidene fluoride-*co*-trifluoroethylene) (P(VDF-TrFE)) nanofibers (Fig. 11e) [113]. Hierarchical micronano morphologies with both beaded and straight nanofibers effectively increased surface roughness and contact areas. Furthermore, interactions between TPU/CB beaded nanofibers and P(VDF-TrFE) nanofibers inhibited P(VDF-TrFE) nanofiber delamination and improved permeability and surface roughness. As a result, electrical performance with a V_{OC} of approximately 49 V and an output power density of 54 mW/m^2, as well as high durability with stable operation for more than 1200 s, was achieved.

Khan et al. reported TENG development with advanced three-dimensional hierarchical microcubic morphological structures employing renewable lignin materials in combination with eco-friendly polyvinyl alcohol (PVA) matrices. The lignin–PVA composite (LPC) films were prepared by solution casting, in which polyvinyl alcohol (PVA) and lignin were crosslinked with epichlorohydrin (EPCH), and subsequently dried at room temperature. The successful creation of this composite from renewable feedstocks highlights the role of sustainable chemical engineering in materials design. Micro- and nanoscale hierarchical

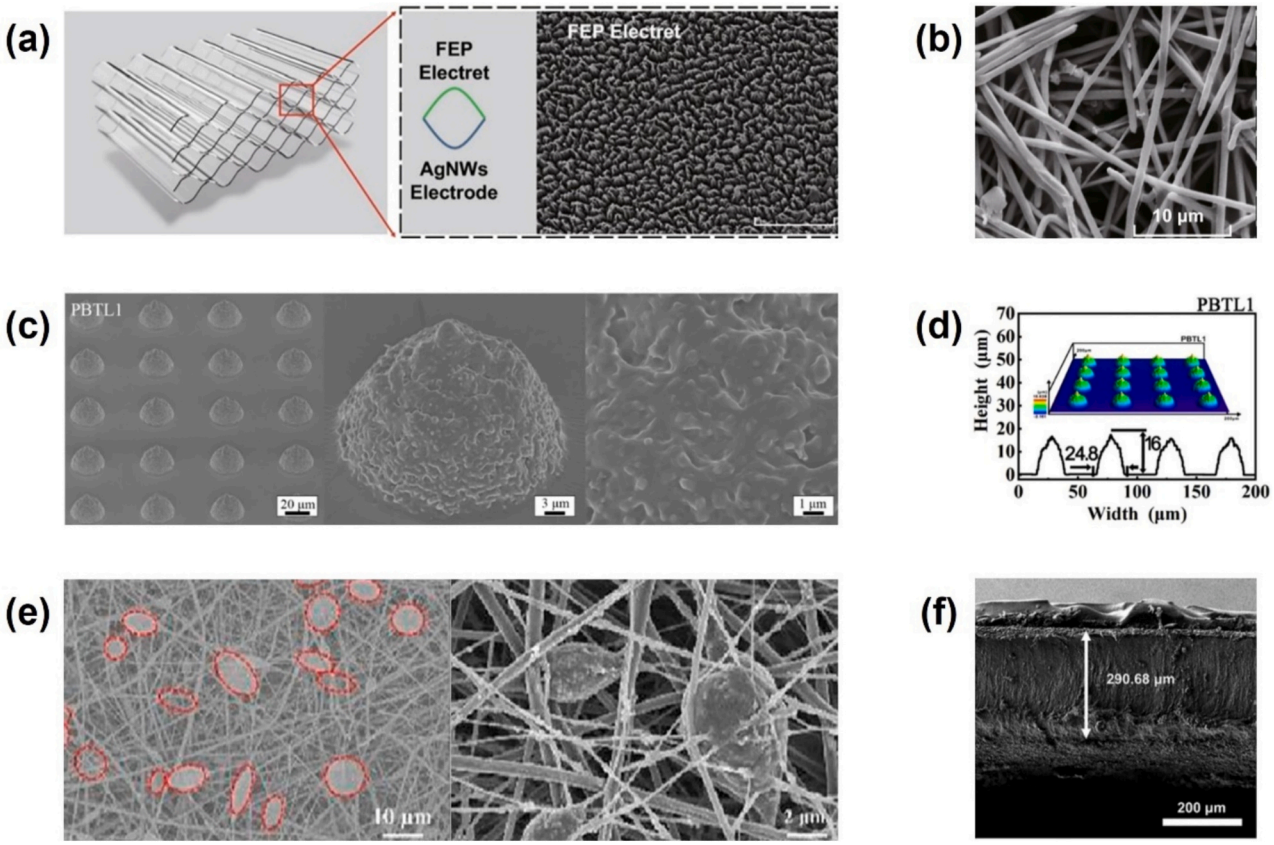


Fig. 11. (a) Schematic of a hierarchical honeycomb-structured electret/triboelectric nanogenerator, and its SEM surface images. (b) SEM image of AgNW-based electrode of a honeycomb-structured energy harvester. Reproduced with permission [112]. Copyright 2023, Springer Nature. (c) SEM images of PBTL1 with various magnification, and (d) its surface morphology. Reproduced with permission [36]. Copyright 2025, American Chemical Society. (e) SEM images of TCP HNMs for TENG material with large-surface area and the enhanced permeability. Reproduced with permission [113]. Copyright 2023, American Chemical Society. (f) Cross-sectional SEM image of micro- and nanoscale hierarchical features on LPC surfaces for enhancing contact areas and interfacial interactions. Reproduced with permission [114]. Copyright 2024, Elsevier B.V.

features on LPC surfaces enhanced contact areas and interfacial interactions, amplifying triboelectric effects. Maximum output voltages of 65 V and output power densities of 135.4 mW/m^2 were achieved. LPC-TENGs established new standards in sensing capabilities and energy harvesting (Fig. 11f) [114]. The analysis of structural engineering approaches confirms that hierarchical designs combining features across multiple length scales represent the most promising strategy for maximizing TENG performance. Microscale structures primarily enhance effective contact areas and mechanical interlocking, while nanoscale features significantly improve charge density and trapping efficiency. The integration of these multi-scale approaches in hierarchical structures achieves synergistic effects that exceed the sum of individual contributions. Bio-inspired designs demonstrate particular promise, offering naturally optimized architectures that balance performance enhancement with mechanical robustness, pointing toward future developments in biomimetic process and material engineering.

5. Research trends and applications

Recent studies have highlighted mechanical stability for long-term operation as critically important [115]. In typical TENG devices, problems including output characteristic degradation caused by surface wear, charge loss, or deformation at electrode-dielectric interfaces have been reported when subjected to repeated external forces or prolonged operation [116]. To overcome these limitations, approaches controlling surface morphology have been utilized to improve device mechanical stability [117]. From a chemical engineering perspective, this challenge is addressed not just by selecting robust materials, but by designing composites and surface structures that effectively manage mechanical stress and dissipate energy, thereby preventing failure. Mechanical stability improvements enhance TENG applicability. In wearable electronics, stable energy harvesting and biosignal detection during skin contact and repeated bending environments became possible. For wearable TENGs, surface structures prioritizing flexibility, durability, and stretchability are most suitable. Examples include microstructured PDMS films with stretchable conductive electrodes, nanofibrous or porous surfaces that provide comfort, and fiber structures that enable multi-mode mechanical deformation (stretching, bending, twisting). These structures maintain conformal skin contact, and mechanical robustness, which are critical for wearable comfort and long-term use. In contrast, large-area harvesters, surface morphologies should emphasize cost-effectiveness, scalability, and ease of fabrication. Simple microstructured arrays created via sandpaper patterning, large-scale fiber coatings, or printable designs enable expansive coverage at low cost. Materials and morphologies that allow roll-to-roll fabrication or 3D printing offer scalability while maintaining reasonable triboelectric performance. The durability demands are less stringent compared to wearables but uniform surface contact and cost efficiency guide morphology choice [49,118,119]. In medical sensors, stable monitoring of body movements or physiological signals enables expansion into diagnostic and healthcare fields [120,121]. Furthermore, in Internet of Things devices, stable power supply maintenance throughout external impacts or environmental changes contributes to long-term self-powered sensor network realization [63]. These successful applications are direct outcomes of advanced material formulation and process engineering that yield devices capable of withstanding real-world operating

conditions. Kang et al. demonstrated self-powered full-color mini LED (mLED) displays enabled by dielectric-tuned triboelectric nanogenerators (DTTENGs). Systems successfully displayed characters including "T," "E," "N," and "G" using blue LEDs without external power supplies. Furthermore, by integrating quantum dot-based color conversion layers, full-color displays with selective control of red, green, and blue emissions were realized (Fig. 12a) [37]. Devices maintained stable RGB operation through complete self-powering, highlighting potential applications in wearable displays, electronic skin, and next-generation IoT platforms.

Wang et al. reported triboelectric tactile sensors (TTS) employing AgNW/NOA electrodes, which produced distinguishable output currents under different contact modes including sliding, tapping, and pushing (Fig. 12b) [38]. During TTS operation, sliding (5 V/70 nA) and tapping (20 V/180 nA) outputs were generated. Notably, high currents of approximately 1000 nA were observed during pushing actions, confirming pronounced electrical responsiveness to mechanical stimuli. These results indicate feasibility of extending TENG-based tactile sensors toward advanced wearable interfaces, medical diagnostic tools, and robotic tactile systems.

Kang et al. fabricated glass-fabric reinforced vinyl-thiol hybrid (GFR-VTH) TENGs, achieving enhanced output performance through polarity modulation via siloxane hybrid matrices (Fig. 12c) [122]. The development of this hybrid matrix is a classic example of polymer and composite engineering. The GFR-VTH TENG operated red, green, blue, and white LEDs, confirming its sufficient power density for practical device applications. These devices were integrated into self-powered body temperature monitoring systems, where TENG-driven capacitors enabled continuous operation of commercial thermometers for real-time axillary temperature measurement. Results underscore capabilities of simple mechanical inputs to sustainably power medical devices, suggesting promise for long-term physiological monitoring without battery replacement requirements.

Lee et al. introduced hierarchical surface texturing strategies in glass-fiber-reinforced siloxane hybrid (SGH) films, incorporating microdomes and nanopores to enhance effective surface areas by approximately 210 % (Fig. 12d) [83]. Resulting THTENGs exhibited nearly four-fold increases in open-circuit voltage and 2.7-fold enhancements in short-circuit current compared to conventional planar counterparts. Importantly, large-area devices (30 × 30 cm^2) reliably powered 750 RGB LEDs in series without external power inputs, underscoring viability for scalable commercial electronic applications. The implementation of hierarchical designs, achieved through multi-step, integrated chemical and physical processing, represents the most promising approach for maximizing triboelectric performance while maintaining structural integrity and mechanical durability.

In addition to TENGs, other energy harvesting technologies, such as piezoelectric, photovoltaic, electromagnetic, and etc. each have their own unique principles and applications. TENGs, based on the combined physical principles of contact electrification and electrostatic induction, remarkable performance in low-frequency environments, a simple structure, and high flexibility, making them suitable for wearables and tactile sensors. Piezoelectric technology converts pressure and deformation into electrical energy, offering stable output and high reliability under repetitive vibrations or momentary pressures, but with limited flexibility [123]. Photovoltaic technology directly converts light into electrical energy, offering the highest power density. However, it is sensitive to indoor and outdoor environmental conditions and changes in illumination, limiting its application to wearables [124]. Electromagnetic technology generates current in response to magnetic field changes, enabling continuous and rapid response. However, it faces limitations in miniaturization, weight reduction, and flexibility, making it advantageous for large devices and high-power systems [125]. Thus, each technology is selected based on given conditions and objectives.

6. Future prospects

The evolutionary trajectory of TENG technology, driven largely by innovations in surface morphology engineering, indicates significant potential for transformative developments across multiple sectors. As the field approaches commercial maturity, future prospects depend on addressing current limitations while capitalizing on emerging

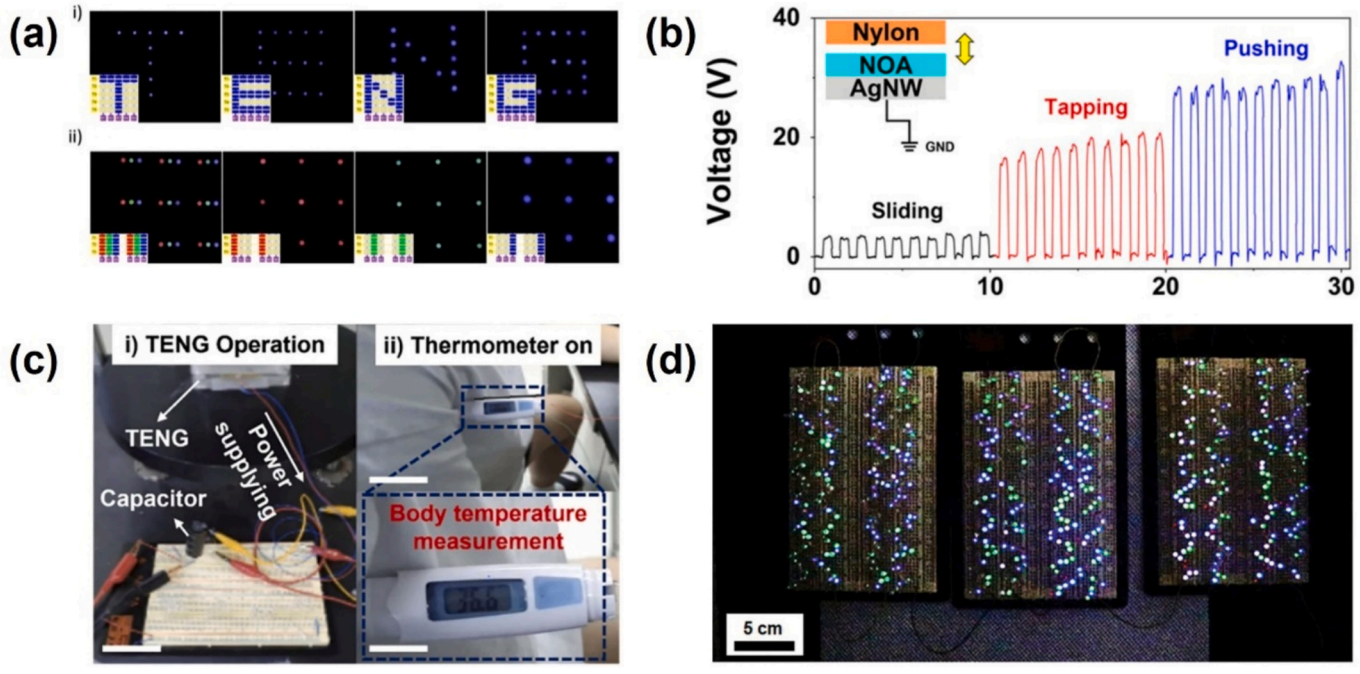


Fig. 12. (a) Photographs of self-powered flexible mLED display without an external power source. Reproduced with permission [37]. Copyright 2021, American Chemical Society. (b) Open-circuit output voltage of SETEH, attached to a human wrist. Reproduced with permission [38]. Copyright 2021, AIP Publishing LLC. (c) Optical images of self-powered body temperature monitoring system. Reproduced with permission [122]. Copyright 2024, Elsevier B.V. (d) 750 RGB LEDs operating image using THTENG without any external power sources. Reproduced with permission [83]. Copyright 2020, John Wiley and Sons.

opportunities in materials science, manufacturing technology, and application domains. This section analyzes the anticipated developments in TENG technology, highlighting the pivotal role of surface engineering in shaping the future landscape of triboelectric energy harvesting systems, as shown in Fig. 13. In 2012, the foundational principles of TENGs were first proposed by Wang's research team, establishing novel energy harvesting concepts [126–128]. This initial phase served as a starting point for subsequent studies focused on enhancing output performance, primarily through fundamental structural and material modifications at the surface level. By 2015, the focus shifted toward wearable electronics, where engineering micro/nanoscale topographies on flexible substrates became critical. By introducing precisely designed structures like pyramids, lines, and pores onto polymers, researchers successfully amplified the effective contact area and charge generation from subtle human movements, demonstrating the first practical self-powered sensors and electronic patches [129–133]. By 2020, TENG technology matured from laboratory concepts to implementation in IoT sensor networks and healthcare monitoring systems. This transition was enabled by advanced surface functionalization and hierarchical structuring, which improved not only power output but also sensitivity and reliability. These engineered surfaces allowed for the stable operation of self-powered wireless sensors and biosignal monitors, solidifying commercialization possibilities [134–142]. Looking ahead to 2025 and beyond, the convergence of TENGs with everyday applications is expected to accelerate, driven by next-generation surface engineering. For textile-based TENGs, the key will be developing scalable chemical coating and surface modification techniques to create durable, hierarchical morphologies on individual fibers, enhancing energy conversion efficiency without compromising comfort or washability [143]. For polymer/nanocomposite devices, engineering the morphology at the material interface will be crucial for creating high-performance, large-area TENGs. This will require breakthroughs in scalable manufacturing processes, such as roll-to-roll nanoimprinting and solution-based coating, which are core competencies of chemical process engineering, to make mass production economically viable.

Despite these promising prospects, the long-term success of TENGs hinges on overcoming key challenges through continued innovation in surface engineering. The durability of nano/micro-scale surface structures against mechanical wear remains a primary concern. Future chemical engineering research must focus on designing wear-resistant hierarchical surfaces or incorporating self-healing functionalities through advanced polymer synthesis. Furthermore, ensuring stable performance under varying environmental conditions, such as humidity, necessitates the development of novel hydrophobic or environmentally adaptive surface morphologies through targeted chemical functionalization.

Despite remarkable advances in the design and performance of TENG systems, several critical challenges remain that must be addressed before their widespread and robust deployment can be realized.

1. Durability and stability concerns

A major issue lies in the long-term mechanical durability and stability of engineered surface structures. Extended cyclic operation can gradually wear and degrade micro/nano-architectures, ultimately leading to significant performance deterioration [144].

2. Environmental and manufacturing constraints

Another important barrier arises from practical operating conditions. Fluctuations in humidity and temperature can induce unpredictable variations in triboelectric output and accelerate material fatigue or failure. At the same time, the absence of scalable and cost-effective manufacturing techniques for large-area, high-throughput production continues to hinder industrial adoption [145].

3. Interfacial degradation mechanisms

Equally critical is the limited understanding of interfacial degradation mechanisms. In hybrid or composite structures, processes such as charge trapping, migration, and chemical instability remain poorly understood, posing unforeseen challenges to practical applications. Addressing these issues will be essential for realizing reliable, high-performance TENGs suitable for commercial and industrial use [146].

Ultimately, surface morphology engineering is the core enabler that



Fig. 13. The evolutionary roadmap of TENG technology driven by surface morphology engineering.
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will bridge the gap between the theoretical potential of TENGs and their widespread practical application. The continued evolution from simple patterned surfaces to multifunctional, intelligent topographies will be the driving force, a progression that relies heavily on advances in chemical and process engineering. Success in realizing these prospects will depend on continued innovation in designing and manufacturing these sophisticated surfaces, determining the extent to which TENGs can fulfill their potential as ubiquitous, sustainable energy harvesting solutions.

7. Conclusion

Surface morphology engineering plays a decisive role in strengthening the performance and broadening the applicability of TENGs. Significant enhancements in charge output, device durability, and miniaturization have been consistently demonstrated through advancements in metal nanostructuring, functional ceramic modifications, and versatile polymer processing. Process integration, especially within the realm of chemical engineering, has become a crucial factor in facilitating synergistic and scalable advancements that surpass the capabilities of individual techniques. Although substantial progress has been achieved, challenges persist, such as long-term mechanical durability and scalability for mass production, that necessitate continued efforts in advanced materials development and sustainable manufacturing. Morphologically engineered TENGs hold significant promise for powering next-generation wearables, self-powered sensor systems, and robust autonomous devices, thereby bridging innovative materials engineering with practical sustainable energy solutions.

CRediT authorship contribution statement

Hee Soo Jin: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Jong Jun Jung:** Visualization, Investigation. **Kang Hyeon Kim:** Validation, Investigation. **Se Jin Choi:** Investigation. **Sang Yoon Park:** Visualization. **Haneol Lee:** Writing – review & editing, Validation, Supervision, Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Regional Innovation System & Education (RISE) program through the Jeonbuk RISE Center, funded by the Ministry of Education (MOE) and the Jeonbuk State, Republic of Korea (2025-RISE-13-JBU), and also supported by the Commercializations Promotion Agency for R&D Outcomes (COMPA) grant funded by the Korean Government (Ministry of Science and ICT, 2023) (RS-2023-00304743).

Data availability

The authors do not have permission to share data.

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